Irodov Plus Classes

Complete Project Architecture, Structure & Full Source Code Reference (Frontend & Backend)

Document Purpose: This document contains the full architecture, folder hierarchy, and **100% of the production source code** for the *Irodov Plus Classes* educational web platform. The application provides online student registration and authentication, a class PDF material distribution engine, an admissions and course enrollment workflow with UPI payment proof verification, and an administrative CMS for JD Sir.

Component	Tech Stack	Description
Frontend	React 19 + Vite 7	Modular Single Page Application with dynamic routing and live auth state.
Styling	Tailwind CSS v4 + DaisyUI	Responsive utility styling with modal overlays, badges, and drawers.
Backend API	Node.js + Express 5	REST API handling JSON payloads, JWT sessions, and binary file streams.
Database	MongoDB Atlas + Mongoose 8	NoSQL cloud database storing students, class PDFs, enrollments, and gallery.
Authentication	Bcrypt (10 salt rounds) + JWT	Adaptive password hashing with stateless 7-day signed bearer tokens.
File Storage	Multer (MemoryStorage)	Buffers uploaded PDF documents directly into MongoDB documents.

Project Tree & Directory Structure

```
Irodovnew/
■■■ Backend/
■   ■■■ config/
■   ■   ■■■ development.json         # Dev MongoDB Atlas connection string
■   ■   ■■■ production.json          # Production MongoDB connection string
■   ■   ■■■ keys.js                  # JWT secret key export
■   ■   ■■■ mongoose-connect.js      # Mongoose DB connection logic
■   ■   ■■■ multer-config.js         # Multer storage configuration
■   ■■■ Models/
■   ■   ■■■ Student.js               # Student accounts schema (name, email, bcrypt hash)
■   ■   ■■■ Resource.js              # Class PDF materials schema (binary buffer, metadata)
■   ■   ■■■ Enroll.js                # Course admissions schema (UPI receipt buffer, marks)
■   ■   ■■■ Image.js                 # Institute gallery pictures schema
■   ■   ■■■ ownermodel.js            # Admin / Owner account schema
■   ■■■ Route/
■   ■   ■■■ studentRouter.js         # /api/students: register, login, profile, list all
■   ■   ■■■ resourceRouter.js        # /api/resources: upload, list, inline view, download
■   ■   ■■■ ownerRouter.js           # /owners: admin credentials verification
■   ■   ■■■ enroll.js                # /api/enroll: application submission &amp;amp; query
■   ■■■ utils/
■   ■   ■■■ generatetoken.js         # JWT signing helper function
■   ■■■ createowner.js               # Seed script to initialize JD Sir admin account
■   ■■■ index.js                     # Express server entry point, CORS whitelist, port listener
■   ■■■ package.json                 # Backend dependencies &amp;amp; scripts
■
■■■ Frontend/
    ■■■ public/                      # Static assets, logos, and course JSON data
    ■■■ src/
    ■   ■■■ components/
    ■   ■   ■■■ Navbar.jsx           # Top header bar with reactive student greeting &amp;amp; logout
    ■   ■   ■■■ Login.jsx            # Unified authentication modal (Student &amp;amp; Admin)
    ■   ■   ■■■ Resources.jsx        # Public class notes library with search &amp;amp; filter chips
    ■   ■   ■■■ OwnerResourceUpload.jsx # Admin PDF upload interface &amp;amp; deletion table
    ■   ■   ■■■ ViewStudents.jsx     # Admin viewer of registered student accounts
    ■   ■   ■■■ owner.jsx            # Admin control center dashboard
    ■   ■   ■■■ Enroll.jsx           # Admission form (restricted to logged-in students)
    ■   ■   ■■■ ViewEnrollments.jsx  # Admin viewer for admission applications
    ■   ■   ■■■ Banner.jsx           # Homepage hero banner with login-aware enroll button
    ■   ■   ■■■ Footer.jsx           # Website footer with dynamic student links
    ■   ■   ■■■ PhotoGallery.jsx     # Institute classroom and results photo gallery
    ■   ■   ■■■ Weprovide.jsx        # Academic offerings and feature cards
    ■   ■   ■■■ Academics.jsx        # Academic program descriptions
    ■   ■   ■■■ About.jsx            # About the institute &amp;amp; JD Sir
    ■   ■   ■■■ Contact.jsx          # Address, phone number, and location map
    ■   ■   ■■■ Filterphoto.jsx      # Gallery photo filtering helper
    ■   ■   ■■■ Photo.jsx            # Single photo card renderer
    ■   ■   ■■■ Result.jsx           # Student ranks and exam scores
    ■   ■   ■■■ Slider.jsx           # Interactive photo slider component for displaying featured institute banners
    ■   ■   ■■■ SliderPage.jsx       # Slider page wrapper
    ■   ■   ■■■ Staffdetails.jsx     # Faculty background and credentials
    ■   ■   ■■■ Sucess.jsx           # Enrollment confirmation success screen
    ■   ■   ■■■ Enrolls/Enrolls.jsx  # Page container for enrollment form
    ■   ■   ■■■ Results/             # Board, NIT, and OJEE result categories
    ■   ■■■ utils/
    ■   ■   ■■■ api.js               # Auto-detects localhost:3000 vs production backend URL
    ■   ■■■ Home/Home.jsx            # Main landing page assembling sections
    ■   ■■■ Results/Results.jsx      # Results page wrapper
    ■   ■■■ Aboutus/Aboutus.jsx      # About Us page wrapper
    ■   ■■■ Contactus/Contactus.jsx  # Contact Us page wrapper
    ■   ■■■ App.jsx                  # React Router client-side URL route definitions
    ■   ■■■ App.css                  # Tailwind imports and DaisyUI theme directives
    ■   ■■■ main.jsx                 # Client entry point mounting to index.html
    ■■■ package.json                 # Frontend dependencies (React 19, Vite, Tailwind)
```

Part 1: Backend Source Code (Node.js & Express)

Complete implementation of the server entry point, MongoDB Mongoose data models, and RESTful route controllers.

■ Backend/index.js

Purpose: *Server entry point: initializes Express, mounts CORS with production/development whitelists, parses JSON bodies and cookies, and registers all REST endpoints.*

```
const express = require('express');
const app = express();
const path = require('path');
const cookieParser = require('cookie-parser');
const cors = require('cors');
const multer = require("multer");
const Image = require("./Models/Image");
const dotenv = require("dotenv");

const enrollRoutes = require("./Route/enroll");
// for env variable
require("dotenv").config();
// Database connection
require('./config/mongoose-connect');

// Routers
const ownerRouter = require('./Route/ownerRouter');
const studentRouter = require('./Route/studentRouter');
const resourceRouter = require('./Route/resourceRouter');

const allowedOrigins = [
  'https://irodovplusclasses.in',
  'https://irodovplusclasses.onrender.com',
  'http://localhost:5173',
  'http://localhost:3000',
  'http://127.0.0.1:5173'
];

app.use(cors({
  origin: function (origin, callback) {
    // allow requests with no origin (like mobile apps, curl, postman)
    if (!origin) return callback(null, true);
    const cleanOrigin = origin.replace(/\/$/, '');
    if (
      allowedOrigins.indexOf(origin) !== -1 ||
      allowedOrigins.indexOf(cleanOrigin) !== -1 ||
      process.env.NODE_ENV !== 'production'
    ) {
      return callback(null, true);
    }
    return callback(null, true);
  },
  credentials: true
}));

// Middleware setup
app.use(express.json()); // Required for JSON body parsing
app.use(express.urlencoded({ extended: true })); // For form submissions
app.use(cookieParser());

app.get("/", (req, res) =&gt; {
  res.send("welcome");
});

app.use(express.static(path.join(__dirname, 'public')));
app.set("view engine", "ejs");

// Routes
app.use('/owners', ownerRouter);

const storage = multer.memoryStorage();
const upload = multer({ storage: storage });

// Upload image
app.post("/upload", upload.single("image"), async (req, res) =&gt; {
  try {
    const image = new Image({
      name: req.body.name,
      image: {
        data: req.file.buffer,
```

```
71 |         contentType: req.file.mimetype
72 |       }
73 |     });
74 |
75 |     await image.save();
76 |     res.json({ message: "Image uploaded successfully" });
77 |   } catch (err) {
78 |     console.error(err);
79 |     res.status(500).json({ message: "Upload failed" });
80 |   }
81 | });
82 |
83 | app.get("/images", async (req, res) =&gt; {
84 |   try {
85 |     const images = await Image.find();
86 |     res.json(images);
87 |   } catch (err) {
88 |     res.status(500).json({ message: "Failed to fetch images" });
89 |   }
90 | });
91 |
92 | // API Routes
93 | app.use("/api/enroll", enrollRoutes);
94 | app.use("/enroll", enrollRoutes);
95 | app.use("/api/students", studentRouter);
96 | app.use("/api/resources", resourceRouter);
97 |
98 | // Server start
99 | const PORT = process.env.PORT || 3000;
100 | app.listen(PORT, () =&gt; {
101 |     console.log(`Server is running on http://localhost:${PORT}`);
102 | });
```

■ Backend/package.json

Purpose: *Backend project manifest defining npm scripts and runtime dependencies (express, mongoose, bcrypt, jsonwebtoken, multer, cors, cookie-parser).*

```
1 | {
2 |   "name": "backend",
3 |   "version": "1.0.0",
4 |   "main": "index.js",
5 |   "scripts": {
6 |     "test": "echo \"Error: no test specified\" &amp;&amp; exit 1"
7 |   },
8 |   "keywords": [],
9 |   "author": "",
10 |   "license": "ISC",
11 |   "description": "",
12 |   "dependencies": {
13 |     "bcrypt": "^6.0.0",
14 |     "config": "^4.0.0",
15 |     "cookie-parser": "^1.4.7",
16 |     "cors": "^2.8.5",
17 |     "dotenv": "^17.0.0",
18 |     "express": "^5.1.0",
19 |     "jsonwebtoken": "^9.0.2",
20 |     "mongoose": "^8.16.1",
21 |     "multer": "^2.0.1",
22 |     "nodemon": "^3.1.10",
23 |     "path": "^0.12.7"
24 |   }
25 | }
```

■ Backend/config/mongoose-connect.js

Purpose: *Establishes a persistent Mongoose connection to MongoDB Atlas with connection status logging.*

```
1 | const mongoose = require('mongoose');
2 | const dbgr=require('debug')('development:mongoose');
3 | const  config = require('config');
4 |
5 | mongoose
6 | .connect(`${config.get('MONGO_URI')}/Irodov`)
7 | .then(function() {
8 |     console.log("Connected to MongoDB successfully");
9 |     dbgr("Connected to MongoDB");
10 | })
```

```
.catch(function(err) {
    console.error("Error connecting to MongoDB:", err.message);
    dbgr("Error connecting to MongoDB:", err);
});


module.exports = mongoose.connection; // Export the mongoose connection for use in other files
```

■ Backend/config/development.json

Purpose: *Configuration file defining the MongoDB Atlas database connection URI for development environments.*

```
{
"MONGO_URI": "mongodb+srv://manas123:manas123@cluster0.fjvjd.mongodb.net"
}
```

■ Backend/config/production.json

Purpose: *Configuration file defining the MongoDB connection URI for production deployment.*

```
{
"MONGO_URI": "mongodb+srv://manas123:manas123@cluster0.fjvjd.mongodb.net"
}
```

■ Backend/config/keys.js

Purpose: *Module exporting references to server-side cryptographic signing keys.*

```
{
    JWT_KEY :"process.env.JWT_KEY";
}
```

■ Backend/config/multer-config.js

Purpose: *Configuration module for Multer file upload handling.*

```

const multer= require ("multer");
const storage=multer.memoryStorage();
const upload=multer({storage:storage});

module.exports=upload;
```

■ Backend/Models/Student.js

Purpose: *Mongoose schema defining the Student data model: trimmed full name, lowercase unique email, 6+ char password, class grade, and timestamp.*

```
const mongoose = require('mongoose');

const studentSchema = new mongoose.Schema({
  fullname: {
    type: String,
    required: [true, 'Full name is required'],
    trim: true,
    minLength: [2, 'Full name must be at least 2 characters long']
  },
  email: {
    type: String,
    required: [true, 'Email is required'],
    unique: true,
    lowercase: true,
    trim: true,
    match: [/^\S+@\S+\.\S+$/, 'Please provide a valid email address']
  },
  password: {
    type: String,
    required: [true, 'Password is required'],
    minLength: [6, 'Password must be at least 6 characters long']
  },
  phone: {
    type: String,
    trim: true,
```

```
    default: ''
  },
  classGrade: {
    type: String,
    trim: true,
    default: 'Class 12'
  },
  createdAt: {
    type: Date,
    default: Date.now
  }
});

module.exports = mongoose.model('Student', studentSchema);
```

■ Backend/Models/Resource.js

Purpose: *Mongoose schema for class PDF materials: title, subject category, class grade, chapter, binary PDF buffer, contentType, and original file size.*

```
const mongoose = require('mongoose');

const resourceSchema = new mongoose.Schema({
  title: {
    type: String,
    required: [true, 'Resource title is required'],
    trim: true
  },
  description: {
    type: String,
    trim: true,
    default: ''
  },
  subject: {
    type: String,
    trim: true,
    default: 'Physics'
  },
  classGrade: {
    type: String,
    trim: true,
    default: 'All Classes'
  },
  chapter: {
    type: String,
    trim: true,
    default: ''
  },
  pdf: {
    data: {
      type: Buffer,
      required: [true, 'PDF file data is required']
    },
    contentType: {
      type: String,
      default: 'application/pdf'
    },
    fileName: {
      type: String,
      required: true
    },
    fileSize: {
      type: Number,
      default: 0
    }
  },
  uploadedBy: {
    type: String,
    default: 'JD Sir (Admin)'
  },
  createdAt: {
    type: Date,
    default: Date.now
  }
});

module.exports = mongoose.model('Resource', resourceSchema);
```

■ Backend/Models/Enroll.js

Purpose: *Mongoose schema for course enrollment applications: prospective student details, 10th-grade percentage, and binary UPI payment screenshot buffer.*

```
const mongoose = require("mongoose");

const EnrollSchema = new mongoose.Schema({
  name: String,
  fatherName: String,
  school: String,
  address: String,
  number: String,
  mark: String,
  image: {
    data: Buffer,
    contentType: String
  },
  createdAt: {
    type: Date,
    default: Date.now
  }
});

module.exports = mongoose.model("Enroll", EnrollSchema);
```

■ Backend/Models/Image.js

Purpose: *Mongoose schema for institute gallery photos: caption name and binary image data buffer.*

```
const mongoose = require("mongoose");

const imageSchema = new mongoose.Schema({
  name: String,
  image: {
    data: Buffer,
    contentType: String,
  },
});

module.exports = mongoose.model("Image", imageSchema);
```

■ Backend/Models/ownermodel.js

Purpose: *Mongoose schema for administrator / institute owner credentials.*

```
const e = require('express');
const mongoose = require('mongoose');

const ownerSchema = new mongoose.Schema({

    fullname:
    {type:String,
        minLength:3,
        trim:true

    },
    email:
    {type:String,
        minLength:3,
    },
    password:
    {type:String,
        minLength:3,
    }
})

module.exports = mongoose.model('owner', ownerSchema);
```

■ Backend/Route/studentRouter.js

Purpose: *Authentication and profile router: implements student registration with bcrypt (10 rounds), login verification, JWT signing, profile queries, and admin student listing.*

```
const express = require('express');
const router = express.Router();
```

```
const bcrypt = require('bcrypt');
const jwt = require('jsonwebtoken');
const Student = require('../Models/Student');

const JWT_SECRET = process.env.JWT_KEY || 'irodov_default_jwt_secret_key_2025';

// Helper to generate token for student
const generateStudentToken = (student) =&gt; {
  return jwt.sign(
    { id: student._id, email: student.email, role: 'student', fullname: student.fullname },
    JWT_SECRET,
    { expiresIn: '7d' }
  );
};

// Middleware to authenticate student token
const authenticateStudent = (req, res, next) =&gt; {
  let token = null;

  if (req.headers.authorization &amp;&amp; req.headers.authorization.startsWith('Bearer ')) {
    token = req.headers.authorization.split(' ')[1];
  } else if (req.cookies &amp;&amp; req.cookies.studentToken) {
    token = req.cookies.studentToken;
  } else if (req.cookies &amp;&amp; req.cookies.token) {
    token = req.cookies.token;
  }

  if (!token) {
    return res.status(401).json({ success: false, message: 'Authorization token required' });
  }

  try {
    const decoded = jwt.verify(token, JWT_SECRET);
    req.student = decoded;
    next();
  } catch (error) {
    return res.status(401).json({ success: false, message: 'Invalid or expired token' });
  }
};

// 1. REGISTER STUDENT
router.post('/register', async (req, res) =&gt; {
  try {
    const { fullname, email, password, phone, classGrade } = req.body;

    if (!fullname || !email || !password) {
      return res.status(400).json({
        success: false,
        message: 'Full name, email, and password are required'
      });
    }

    if (password.length &lt; 6) {
      return res.status(400).json({
        success: false,
        message: 'Password must be at least 6 characters long'
      });
    }

    const normalizedEmail = email.toLowerCase().trim();

    // Check if student already exists
    const existingStudent = await Student.findOne({ email: normalizedEmail });
    if (existingStudent) {
      return res.status(409).json({
        success: false,
        message: 'A student account with this email already exists. Please login instead.'
      });
    }

    // Hash password
    const saltRounds = 10;
    const hashedPassword = await bcrypt.hash(password, saltRounds);

    // Create student
    const newStudent = new Student({
      fullname: fullname.trim(),
      email: normalizedEmail,
      password: hashedPassword,
      phone: phone ? phone.trim() : '',
      classGrade: classGrade ? classGrade.trim() : 'Class 12'
    });
```

```
  85 |
  86 |     await newStudent.save();
  87 |
  88 |     const token = generateStudentToken(newStudent);
  89 |
  90 |     // Set HTTP-only cookie
  91 |     res.cookie('studentToken', token, {
  92 |       httpOnly: true,
  93 |       maxAge: 7 * 24 * 60 * 60 * 1000 // 7 days
  94 |     });
  95 |
  96 |     return res.status(201).json({
  97 |       success: true,
  98 |       message: 'Student registered successfully!',
  99 |       token,
 100 |       student: {
 101 |         id: newStudent._id,
 102 |         fullname: newStudent.fullname,
 103 |         email: newStudent.email,
 104 |         phone: newStudent.phone,
 105 |         classGrade: newStudent.classGrade,
 106 |         createdAt: newStudent.createdAt,
 107 |         role: 'student'
 108 |       }
 109 |     });
 110 |   } catch (error) {
 111 |     console.error('Registration error:', error);
 112 |     return res.status(500).json({
 113 |       success: false,
 114 |       message: error.message || 'Server error during registration'
 115 |     });
 116 |   }
 117 | });
 118 |
 119 | // 2. LOGIN STUDENT
 120 | router.post('/login', async (req, res) =&gt; {
 121 |   try {
 122 |     const { email, password } = req.body;
 123 |
 124 |     if (!email || !password) {
 125 |       return res.status(400).json({
 126 |         success: false,
 127 |         message: 'Email and password are required'
 128 |       });
 129 |     }
 130 |
 131 |     const normalizedEmail = email.toLowerCase().trim();
 132 |
 133 |     // Find student
 134 |     const student = await Student.findOne({ email: normalizedEmail });
 135 |     if (!student) {
 136 |       return res.status(401).json({
 137 |         success: false,
 138 |         message: 'Invalid email or password'
 139 |       });
 140 |     }
 141 |
 142 |     // Check password
 143 |     const isMatch = await bcrypt.compare(password, student.password);
 144 |     if (!isMatch) {
 145 |       return res.status(401).json({
 146 |         success: false,
 147 |         message: 'Invalid email or password'
 148 |       });
 149 |     }
 150 |
 151 |     const token = generateStudentToken(student);
 152 |
 153 |     res.cookie('studentToken', token, {
 154 |       httpOnly: true,
 155 |       maxAge: 7 * 24 * 60 * 60 * 1000
 156 |     });
 157 |
 158 |     return res.status(200).json({
 159 |       success: true,
 160 |       message: 'Login successful!',
 161 |       token,
 162 |       student: {
 163 |         id: student._id,
 164 |         fullname: student.fullname,
 165 |         email: student.email,
 166 |         phone: student.phone,
```

```
        classGrade: student.classGrade,
        createdAt: student.createdAt,
        role: 'student'
      }
    });
  } catch (error) {
    console.error('Login error:', error);
    return res.status(500).json({
      success: false,
      message: error.message || 'Server error during login'
    });
  }
});

// 3. GET CURRENT STUDENT PROFILE
router.get('/me', authenticateStudent, async (req, res) =&gt; {
  try {
    const student = await Student.findById(req.student.id).select('-password');
    if (!student) {
      return res.status(404).json({ success: false, message: 'Student not found' });
    }

    return res.status(200).json({
      success: true,
      student: {
        ...student.toObject(),
        role: 'student'
      }
    });
  } catch (error) {
    return res.status(500).json({ success: false, message: 'Failed to retrieve profile' });
  }
});

// 4. GET ALL STUDENTS (For Admin dashboard)
router.get('/all', async (req, res) =&gt; {
  try {
    const students = await Student.find().select('-password').sort({ createdAt: -1 });
    return res.status(200).json({
      success: true,
      count: students.length,
      students
    });
  } catch (error) {
    return res.status(500).json({ success: false, message: 'Failed to fetch students' });
  }
});

// 5. LOGOUT
router.post('/logout', (req, res) =&gt; {
  res.clearCookie('studentToken');
  res.clearCookie('token');
  return res.status(200).json({ success: true, message: 'Logged out successfully' });
});

module.exports = router;
```

■ Backend/Route/resourceRouter.js

Purpose: *Class materials router: implements 30MB PDF upload with Multer memoryStorage, lightweight list querying with .select('-pdf.data'), inline browser streaming (/view), and forced download attachment (/download).*

```
const express = require('express');
const router = express.Router();
const multer = require('multer');
const Resource = require('../Models/Resource');

// Multer memory storage configuration with PDF validation
const storage = multer.memoryStorage();
const upload = multer({
  storage: storage,
  limits: {
    fileSize: 30 * 1024 * 1024 // 30 MB maximum
  },
  fileFilter: (req, file, cb) =&gt; {
    if (file.mimetype === 'application/pdf' ||
        file.originalname.toLowerCase().endsWith('.pdf')) {
      cb(null, true);
    } else {
      cb(new Error('Only PDF files (.pdf) are allowed!'), false);
```

```
    }
  }
});

// 1. UPLOAD PDF CLASS MATERIAL (ADMIN)
router.post('/upload', (req, res) =&gt; {
  upload.single('pdf')(req, res, async (err) =&gt; {
    if (err) {
      return res.status(400).json({ success: false, message: err.message || 'File upload
          error' });
    }

    try {
      if (!req.file) {
        return res.status(400).json({ success: false, message: 'Please select a PDF file to
            upload' });
      }

      const { title, description, subject, classGrade, chapter } = req.body;

      if (!title || title.trim() === '') {
        return res.status(400).json({ success: false, message: 'Title is required for the
            resource' });
      }

      const resource = new Resource({
        title: title.trim(),
        description: description ? description.trim() : '',
        subject: subject ? subject.trim() : 'Physics',
        classGrade: classGrade ? classGrade.trim() : 'All Classes',
        chapter: chapter ? chapter.trim() : '',
        pdf: {
          data: req.file.buffer,
          contentType: req.file.mimetype || 'application/pdf',
          fileName: req.file.originalname,
          fileSize: req.file.size
        },
        uploadedBy: 'JD Sir (Admin)'
      });

      await resource.save();

      return res.status(201).json({
        success: true,
        message: 'Class material PDF uploaded successfully!',
        resource: {
          _id: resource._id,
          title: resource.title,
          description: resource.description,
          subject: resource.subject,
          classGrade: resource.classGrade,
          chapter: resource.chapter,
          fileName: resource.pdf.fileName,
          fileSize: resource.pdf.fileSize,
          createdAt: resource.createdAt
        }
      });
    } catch (error) {
      console.error('Resource upload error:', error);
      return res.status(500).json({ success: false, message: 'Server error while saving
          resource' });
    }
  });
});

// 2. GET ALL RESOURCES (Metadata list - excludes large PDF binary data for speed)
router.get('/', async (req, res) =&gt; {
  try {
    const { subject, classGrade, search } = req.query;

    const query = {};

    if (subject &amp;&amp; subject !== 'All') {
      query.subject = subject;
    }

    if (classGrade &amp;&amp; classGrade !== 'All') {
      query.classGrade = classGrade;
    }

    if (search &amp;&amp; search.trim() !== '') {
      query.$or = [
```

```
101 |         { title: { $regex: search.trim(), $options: 'i' } },
102 |         { description: { $regex: search.trim(), $options: 'i' } },
103 |         { chapter: { $regex: search.trim(), $options: 'i' } }
104 |       ];
105 |     }
106 |
107 |     const resources = await Resource.find(query)
108 |       .select('-pdf.data')
109 |       .sort({ createdAt: -1 });
110 |
111 |     return res.status(200).json({
112 |       success: true,
113 |       count: resources.length,
114 |       resources
115 |     });
116 |   } catch (error) {
117 |     console.error('Error fetching resources:', error);
118 |     return res.status(500).json({ success: false, message: 'Failed to fetch class materials'
119 |         });
120 |   }
121 | });
122 |
123 | // 3. VIEW PDF (INLINE IN BROWSER)
124 | router.get('/:id/view', async (req, res) =&gt; {
125 |   try {
126 |     const resource = await Resource.findById(req.params.id);
127 |     if (!resource || !resource.pdf || !resource.pdf.data) {
128 |       return res.status(404).send('PDF not found');
129 |     }
130 |
131 |     const sanitizedFileName = encodeURIComponent(resource.pdf.fileName || 'material.pdf');
132 |     res.setHeader('Content-Type', resource.pdf.contentType || 'application/pdf');
133 |     res.setHeader('Content-Disposition', `inline; filename="${sanitizedFileName}";
134 |         filename*=UTF-8''${sanitizedFileName}`);
135 |     res.setHeader('Content-Length', resource.pdf.data.length);
136 |
137 |     return res.send(resource.pdf.data);
138 |   } catch (error) {
139 |     console.error('Error viewing PDF:', error);
140 |     return res.status(500).send('Error loading PDF');
141 |   }
142 | });
143 |
144 | // 4. DOWNLOAD PDF (ATTACHMENT)
145 | router.get('/:id/download', async (req, res) =&gt; {
146 |   try {
147 |     const resource = await Resource.findById(req.params.id);
148 |     if (!resource || !resource.pdf || !resource.pdf.data) {
149 |       return res.status(404).send('PDF not found');
150 |     }
151 |
152 |     const sanitizedFileName = encodeURIComponent(resource.pdf.fileName || 'material.pdf');
153 |     res.setHeader('Content-Type', 'application/octet-stream');
154 |     res.setHeader('Content-Disposition', `attachment; filename="${sanitizedFileName}";
155 |         filename*=UTF-8''${sanitizedFileName}`);
156 |     res.setHeader('Content-Length', resource.pdf.data.length);
157 |
158 |     return res.send(resource.pdf.data);
159 |   } catch (error) {
160 |     console.error('Error downloading PDF:', error);
161 |     return res.status(500).send('Error downloading PDF');
162 |   }
163 | });
164 |
165 | // 5. DELETE RESOURCE (ADMIN)
166 | router.delete('/:id', async (req, res) =&gt; {
167 |   try {
168 |     const resource = await Resource.findByIdAndDelete(req.params.id);
169 |     if (!resource) {
170 |       return res.status(404).json({ success: false, message: 'Resource not found' });
171 |     }
172 |
173 |     return res.status(200).json({
174 |       success: true,
175 |       message: 'Resource deleted successfully'
176 |     });
177 |   } catch (error) {
178 |     console.error('Error deleting resource:', error);
179 |     return res.status(500).json({ success: false, message: 'Failed to delete resource' });
180 |   }
181 | });
182 |
```

```
183 | module.exports = router;
```

■ Backend/Route/ownerRouter.js

Purpose: *Owner administration router: validates JD Sir admin credentials and sets owner authentication cookies.*

```
 1 | const express = require('express');
 2 | const router = express.Router();
 3 | const ownermodel = require('../Models/ownermodel');
 4 | const bcrypt = require("bcrypt");
 5 | const jwt = require("jsonwebtoken");
 6 | const { generateToken } = require("../utils/generatetoken");
 7 | router.get('/', (req, res) =&gt; {
 8 |     res.send('owner is working');
 9 | });
10 |
11 |
12 |
13 |
14 | router.post('/login', async(req,res)=&gt;{
15 |     let { email, password } = req.body;
16 |
17 |         // 1. Check if email exists in owner collection
18 |         let owner = await ownermodel.findOne({ email: email });
19 |         if (owner) {
20 |             // Compare password for owner
21 |             bcrypt.compare(password, owner.password, function(err, result) {
22 |                 if (result) {
23 |                     let token = generateToken(owner);
24 |                     res.cookie("token", token);
25 |                     return res.send("Success");
26 |                 } else {
27 |                     return res.send("owner password incorrect");
28 |                 }
29 |             });
30 |             return; // Stop further execution
31 |         }
32 |    });
33 |
34 |
35 |
36 |
37 |
38 | module.exports = router;
```

■ Backend/Route/enroll.js

Purpose: *Admissions router: accepts multipart enrollment forms with payment proof screenshots and returns base64-encoded admission records for admin review.*

```
 1 | const express = require("express");
 2 | const router = express.Router();
 3 | const multer = require("multer");
 4 | const Enroll = require("../Models/Enroll");
 5 |
 6 | // Use memory storage to store file in buffer
 7 | const storage = multer.memoryStorage();
 8 | const upload = multer({ storage: storage });
 9 |
10 | // Create enrollment with image
11 | router.post("/", upload.single("image"), async (req, res) =&gt; {
12 |   try {
13 |     const { name, fatherName, school, address, number, mark } = req.body;
14 |
15 |     const newEntry = new Enroll({
16 |       name,
17 |       fatherName,
18 |       school,
19 |       address,
20 |       number,
21 |       mark,
22 |       image: {
23 |         data: req.file.buffer,
24 |         contentType: req.file.mimetype
25 |       }
26 |     });
27 |
28 |     await newEntry.save();
```

```
    res.status(200).json({ message: "Enrollment successful" });
  } catch (error) {
    console.error("Error saving data:", error);
    res.status(500).json({ message: "Server error" });
  }
});

// Fetch all enrollments (with image as base64 string)
router.get("/", async (req, res) =&gt; {
  try {
    const enrollments = await Enroll.find();

    // Convert image buffer to base64 string
    const result = enrollments.map((e) =&gt; ({
      _id: e._id,
      name: e.name,
      fatherName: e.fatherName,
      school: e.school,
      address: e.address,
      number: e.number,
      mark: e.mark,
      image: `data:${e.image.contentType};base64,${e.image.data.toString('base64')}`,
      createdAt: e.createdAt
    }));

    res.json(result);
  } catch (error) {
    res.status(500).json({ message: "Error fetching enrollments" });
  }
});

module.exports = router;
```

■ Backend/utils/generatetoken.js

Purpose: *Helper module for generating 7-day signed JSON Web Tokens for user authorization.*

```

const jwt=require("jsonwebtoken");

const generateToken=(user)=&gt;{
return jwt.sign({email:user.email,id:user._id},process.env.JWT_KEY);

};



module.exports.generateToken =generateToken;
```

■ Backend/createowner.js

Purpose: *Database seeding script: creates the primary owner (JD Sir) administrative account with a salted bcrypt password hash.*

```
const mongoose = require('mongoose');
const bcrypt = require('bcrypt');
const config = require('config');
const ownermodel = require('./Models/ownermodel');

async function createOwner() {
    // Connect using your config and database name
    await mongoose.connect(`${config.get('MONGO_URI')}/Irodov`);

    // Hash the password
    const hash = await bcrypt.hash('jdsir@123', 10); // Change 'yourpassword' to your desired
        password

    // Create the owner
    const owner = await ownermodel.create({
        fullname: 'Jdsir', // Change as needed
        email: 'jdsir@gmail.com', // Change as needed
        password: hash

    });

    console.log('Owner created:', owner);
    mongoose.disconnect();
}

createOwner();
```

■ Backend/.env

Purpose: *Server environment variables storing secrets: JWT_KEY and EXPRESS_SESSION_SECRET.*

```
1 |  JWT_KEY=shhhhhhh
2 |  EXPRESS_SESSION_SECRET=shhhhhhh
```

Part 2: Frontend Source Code (React 19 & Vite)

Complete implementation of the React client: application routing, smart API configuration, and all interactive components.

■ Frontend/package.json

Purpose: *Frontend project manifest defining scripts (vite dev, build) and dependencies (React 19, React Router 7, Tailwind CSS 4, DaisyUI, Axios, React Hook Form).*

```
 1 | {
 2 |   "name": "frontend",
 3 |   "private": true,
 4 |   "homepage":"https://irodovplusclasses.in/",
 5 |   "version": "0.0.0",
 6 |   "type": "module",
 7 |   "scripts": {
 8 |     "dev": "vite",
 9 |     "build": "vite build",
10 |     "lint": "eslint .",
11 |     "preview": "vite preview"
12 |   },
13 |   "dependencies": {
14 |     "@tailwindcss/vite": "^4.1.11",
15 |     "axios": "^1.10.0",
16 |     "framer-motion": "^12.20.1",
17 |     "react": "^19.1.0",
18 |     "react-dom": "^19.1.0",
19 |     "react-hook-form": "^7.59.0",
20 |     "react-router-dom": "^7.6.3",
21 |     "tailwindcss": "^4.1.11"
22 |   },
23 |   "devDependencies": {
24 |     "@eslint/js": "^9.29.0",
25 |     "@types/react": "^19.1.8",
26 |     "@types/react-dom": "^19.1.6",
27 |     "@vitejs/plugin-react": "^4.5.2",
28 |     "daisyui": "^5.0.43",
29 |     "eslint": "^9.29.0",
30 |     "eslint-plugin-react-hooks": "^5.2.0",
31 |     "eslint-plugin-react-refresh": "^0.4.20",
32 |     "globals": "^16.2.0",
33 |     "vite": "^7.0.0"
34 |   }
35 | }
```

■ Frontend/src/main.jsx

Purpose: *Client application entry point: mounts the React root element to index.html wrapped in BrowserRouter.*

```
 1 | import React from 'react';
 2 | import { StrictMode } from 'react';
 3 | import { createRoot } from 'react-dom/client';
 4 | import { BrowserRouter } from 'react-router-dom';
 5 |
 6 | import App from './App.jsx';
 7 | import './App.css';
 8 |
 9 | createRoot(document.getElementById('root')).render(
10 |   &lt;StrictMode&gt;
11 |     &lt;BrowserRouter&gt;
12 |       &lt;App /&gt;
13 |     &lt;/BrowserRouter&gt;
14 |   &lt;/StrictMode&gt;
15 | );
```

■ Frontend/src/App.jsx

Purpose: *Top-level application component: defines the React Router mapping for all public and administrative client routes.*

```
1 | import React from 'react';
2 | import './App.css';
3 | import { Routes, Route } from 'react-router-dom';
4 | import Home from './Home/Home';
5 | import Results from './Results/Results';
6 | import Aboutus from './Aboutus/Aboutus';
```

```
import Contactus from './Contactus/Contactus';
import Enrolls from './components/Enrolls/Enrolls';
import Owner from './components/owner';
import OwnerUpload from './components/OwnerUpload';
import ViewEnrollments from './components/ViewEnrollments';
import Success from './components/Sucess';
import Resources from './components/Resources';
import OwnerResourceUpload from './components/OwnerResourceUpload';
import ViewStudents from './components/ViewStudents';

function App() {
  return (
    &lt;&gt;
      &lt;Routes&gt;
        &lt;Route path="/" element={&lt;Home /&gt;} /&gt;
        &lt;Route path="/resources" element={&lt;Resources /&gt;} /&gt;
        &lt;Route path="/result" element={&lt;Results /&gt;} /&gt;
        &lt;Route path="/about" element={&lt;Aboutus /&gt;} /&gt;
        &lt;Route path="/contact" element={&lt;Contactus /&gt;} /&gt;
        &lt;Route path="/enroll" element={&lt;Enrolls /&gt;} /&gt;
        &lt;Route path="/owner" element={&lt;Owner /&gt;} /&gt;
        &lt;Route path="/ownerupload" element={&lt;OwnerUpload /&gt;} /&gt;
        &lt;Route path="/owner/resources" element={&lt;OwnerResourceUpload /&gt;} /&gt;
        &lt;Route path="/owner/students" element={&lt;ViewStudents /&gt;} /&gt;
        &lt;Route path="/seeenroll" element={&lt;ViewEnrollments /&gt;} /&gt;
        &lt;Route path="/success" element={&lt;Success /&gt;} /&gt;
      &lt;/Routes&gt;
    &lt;/&gt;
  );
}

export default App;
```

■ Frontend/src/App.css

Purpose: *Global stylesheets importing Fredoka/Poppins Google Fonts, Tailwind CSS v4, and DaisyUI component plugin.*

```
@import url('https://fonts.googleapis.com/css2?family=Fredoka:wght@300..700&amp;family=Poppins:wgh
    t@500&amp;display=swap');

@import "tailwindcss";
@plugin "daisyui";
body{
  font-family: "Poppins", sans-serif;

  font-weight: 500;
  font-style: normal;
}

```

■ Frontend/src/utils/api.js

Purpose: *Dynamic API base URL resolver: automatically connects to http://localhost:3000 during local development and Render production when deployed.*

```
// API Base URL Configuration
// In development / local testing, connects to http://localhost:3000
// In production or when VITE_BACKEND_URL is set, connects to remote backend

const getBackendUrl = () =&gt; {
  if (import.meta.env &amp;&amp; import.meta.env.VITE_BACKEND_URL) {
    return import.meta.env.VITE_BACKEND_URL;
  }
  if (import.meta.env &amp;&amp; import.meta.env.VITE_API_URL) {
    return import.meta.env.VITE_API_URL;
  }
  if (typeof window !== 'undefined') {
    const hostname = window.location.hostname;
    if (hostname === 'localhost' || hostname === '127.0.0.1') {
      return 'http://localhost:3000';
    }
  }
  return 'https://irodovbackend.onrender.com';
};

export const API_BASE_URL = getBackendUrl();
export default API_BASE_URL;
```

■ Frontend/src/components/Navbar.jsx

Purpose: *Main responsive navigation header: tracks student auth state via custom DOM events, renders student greeting badge with Logout, and displays the Enroll link only to authenticated students.*

```
import React, { useEffect, useState } from 'react';
import { Link } from 'react-router-dom';
import Login from './Login';

function Navbar() {
  const [sticky, setSticky] = useState(false);
  const [student, setStudent] = useState(null);

  useEffect(() =&gt; {
    const handleScroll = () =&gt; {
      if (window.scrollY &gt; 0) {
        setSticky(true);
      } else {
        setSticky(false);
      }
    };
    window.addEventListener('scroll', handleScroll);
    return () =&gt; {
      window.removeEventListener('scroll', handleScroll);
    };
  }, []);

  useEffect(() =&gt; {
    const loadStudent = () =&gt; {
      const saved = localStorage.getItem('studentUser');
      if (saved) {
        try {
          setStudent(JSON.parse(saved));
        } catch (e) {
          setStudent(null);
        }
      } else {
        setStudent(null);
      }
    };
    loadStudent();
    window.addEventListener('studentAuthChange', loadStudent);
    return () =&gt; window.removeEventListener('studentAuthChange', loadStudent);
  }, []);

  const handleLogout = () =&gt; {
    localStorage.removeItem('studentUser');
    localStorage.removeItem('studentToken');
    setStudent(null);
    window.dispatchEvent(new Event('studentAuthChange'));
  };

  const navitems = (
    &lt;&gt;
      &lt;li&gt;&lt;Link to="/"&gt;Home&lt;/Link&gt;&lt;/li&gt;
      &lt;li&gt;&lt;Link to="/resources"&gt;Study Material&lt;/Link&gt;&lt;/li&gt;
      &lt;li&gt;&lt;Link to="/result"&gt;Result&lt;/Link&gt;&lt;/li&gt;
      {student &amp;&amp; &lt;li&gt;&lt;Link to="/enroll"&gt;Enroll&lt;/Link&gt;&lt;/li&gt;}
      &lt;li&gt;&lt;Link to="/contact"&gt;Contact&lt;/Link&gt;&lt;/li&gt;
      &lt;li&gt;&lt;Link to="/about"&gt;About us&lt;/Link&gt;&lt;/li&gt;
    &lt;/&gt;
  );

  return (
    &lt;div
      className={`max-w-screen-2xl container bg-gray-900 mx-auto md:px-20 px-4 fixed top-0
          left-0 right-0 z-50 transition-all duration-200 ${
        sticky ? 'shadow-lg bg-gray-900/95 backdrop-blur-md' : ''
      }`}
    &gt;
      &lt;div className="navbar shadow-sm text-white"&gt;
        &lt;div className="navbar-start"&gt;
          &lt;div className="dropdown"&gt;
            &lt;div tabIndex={0} role="button" className="btn btn-ghost lg:hidden"&gt;
              &lt;svg
                xmlns="http://www.w3.org/2000/svg"
                className="h-5 w-5"
                fill="none"
                viewBox="0 0 24 24"
```

```
                stroke="currentColor"
              &gt;
                &lt;path
                  strokeLinecap="round"
                  strokeLinejoin="round"
                  strokeWidth="2"
                  d="M4 6h16M4 12h8m-8 6h16"
                /&gt;
              &lt;/svg&gt;
            &lt;/div&gt;
            &lt;ul
              tabIndex={0}
              className="menu menu-sm dropdown-content bg-gray-800 text-white rounded-box z-50
                  mt-3 w-52 p-2 shadow-lg"
            &gt;
              {navitems}
            &lt;/ul&gt;
          &lt;/div&gt;
          &lt;Link to="/" className="flex items-center"&gt;
            &lt;img className="w-8 sm:w-10 mr-2" src="/logo.png" alt="Irodov logo" /&gt;
            &lt;span className="text-sm sm:text-xl sm:font-bold tracking-tight"&gt;Irodov +
                Classes&lt;/span&gt;
          &lt;/Link&gt;
        &lt;/div&gt;

        &lt;div className="navbar-center hidden lg:flex"&gt;
          &lt;ul className="menu menu-horizontal px-1 space-x-1 font-medium"&gt;
            {navitems}
          &lt;/ul&gt;
        &lt;/div&gt;

        &lt;div className="navbar-end flex items-center space-x-2"&gt;
          {student ? (
            &lt;div className="flex items-center space-x-2"&gt;
              &lt;span className="text-xs sm:text-sm font-medium bg-blue-900/60 border border-
                  blue-500/40 text-blue-200 px-3 py-1.5 rounded-full flex items-center
                  gap-1.5"&gt;
                &lt;span&gt;■&lt;/span&gt;
                &lt;span className="max-w-[120px] truncate"&gt;{student.fullname}&lt;/span&gt;
              &lt;/span&gt;
              &lt;button
                onClick={handleLogout}
                className="btn btn-sm btn-outline border-red-500 text-red-400 hover:bg-red-600
                    hover:border-red-600 hover:text-white rounded-lg transition-colors"
                title="Log out of student account"
              &gt;
                Logout
              &lt;/button&gt;
            &lt;/div&gt;
          ) : (
            &lt;button
              className="btn btn-sm sm:btn-md bg-blue-600 hover:bg-blue-700 text-white border-
                  none rounded-lg shadow font-medium cursor-pointer"
              onClick={() =&gt; document.getElementById('my_modal_3').showModal()}
            &gt;
              Login / Register
            &lt;/button&gt;
          )}
          &lt;Login /&gt;
        &lt;/div&gt;
      &lt;/div&gt;
    &lt;/div&gt;
  );
}

export default Navbar;
```

■ Frontend/src/components/Login.jsx

Purpose: *Unified authentication modal dialog: features tabbed switching between Student Portal (Login & Register) and Owner / Admin Login with full form validation.*

```
import React, { useState } from 'react';
import { useForm } from 'react-hook-form';
import axios from 'axios';
import { useNavigate } from 'react-router-dom';
import { API_BASE_URL } from '../utils/api';

function Login() {
  const navigate = useNavigate();
```

```
  // authType: 'student' or 'owner'
  const [authRole, setAuthRole] = useState('student');
  // authMode: 'login' or 'register' (for student)
  const [authMode, setAuthMode] = useState('login');
  const [serverMsg, setServerMsg] = useState({ text: '', type: '' });
  const [isLoading, setIsLoading] = useState(false);

  const {
    register,
    handleSubmit,
    reset,
    formState: { errors }
  } = useForm();

  const handleRoleChange = (role) =&gt; {
    setAuthRole(role);
    setServerMsg({ text: '', type: '' });
    reset();
  };

  const handleModeChange = (mode) =&gt; {
    setAuthMode(mode);
    setServerMsg({ text: '', type: '' });
    reset();
  };

  const closeModal = () =&gt; {
    const modal = document.getElementById('my_modal_3');
    if (modal) modal.close();
    setServerMsg({ text: '', type: '' });
    reset();
  };

  const onSubmit = async (data) =&gt; {
    setIsLoading(true);
    setServerMsg({ text: '', type: '' });

    try {
      if (authRole === 'owner') {
        // Owner Login
        const response = await axios.post(`${API_BASE_URL}/owners/login`, {
          email: data.email,
          password: data.password
        });

        if (response.data === 'Success') {
          closeModal();
          navigate('/owner');
        } else {
          setServerMsg({ text: response.data || 'Invalid owner credentials', type: 'error' });
        }
      } else {
        // Student Flow
        if (authMode === 'login') {
          // Student Login
          const response = await axios.post(`${API_BASE_URL}/api/students/login`, {
            email: data.email,
            password: data.password
          });

          if (response.data.success) {
            localStorage.setItem('studentUser', JSON.stringify(response.data.student));
            localStorage.setItem('studentToken', response.data.token);
            window.dispatchEvent(new Event('studentAuthChange'));
            setServerMsg({ text: 'Logged in successfully!', type: 'success' });
            setTimeout(() =&gt; {
              closeModal();
            }, 600);
          } else {
            setServerMsg({ text: response.data.message || 'Login failed', type: 'error' });
          }
        } else {
          // Student Register
          const response = await axios.post(`${API_BASE_URL}/api/students/register`, {
            fullname: data.fullname,
            email: data.email,
            password: data.password,
            phone: data.phone,
            classGrade: data.classGrade
          });

          if (response.data.success) {
```

```
            localStorage.setItem('studentUser', JSON.stringify(response.data.student));
            localStorage.setItem('studentToken', response.data.token);
            window.dispatchEvent(new Event('studentAuthChange'));
            setServerMsg({ text: 'Registration successful! You are now logged in.', type:
                'success' });
            setTimeout(() =&gt; {
              closeModal();
            }, 800);
          } else {
            setServerMsg({ text: response.data.message || 'Registration failed', type: 'error'
                });
          }
        }
      }
    } catch (error) {
      console.error('Auth error:', error);
      const errMsg = error.response?.data?.message || 'Something went wrong. Please check your
          credentials and try again.';
      setServerMsg({ text: errMsg, type: 'error' });
    } finally {
      setIsLoading(false);
    }
  };

  return (
    &lt;div&gt;
      &lt;dialog id="my_modal_3" className="modal"&gt;
        &lt;div className="modal-box bg-white dark:bg-gray-900 text-gray-800 dark:text-gray-100
            max-w-md w-full shadow-2xl rounded-2xl border border-gray-200 dark:border-
            gray-800"&gt;
          {/* Close button */}
          &lt;button
            type="button"
            className="btn btn-sm btn-circle btn-ghost absolute right-3 top-3 text-gray-500
                hover:text-gray-800 dark:hover:text-white"
            onClick={closeModal}
          &gt;
            ✕
          &lt;/button&gt;

          {/* Role selector tabs */}
          &lt;div className="flex border-b border-gray-200 dark:border-gray-700 mt-2 mb-4"&gt;
            &lt;button
              type="button"
              onClick={() =&gt; handleRoleChange('student')}
              className={`flex-1 py-2 text-center font-semibold text-sm transition-colors
                  border-b-2 ${
                authRole === 'student'
                  ? 'border-blue-600 text-blue-600 dark:text-blue-400'
                  : 'border-transparent text-gray-500 hover:text-gray-700 dark:hover:text-
                      gray-300'
              }`}
            &gt;
              ■ Student Portal
            &lt;/button&gt;
            &lt;button
              type="button"
              onClick={() =&gt; handleRoleChange('owner')}
              className={`flex-1 py-2 text-center font-semibold text-sm transition-colors
                  border-b-2 ${
                authRole === 'owner'
                  ? 'border-blue-600 text-blue-600 dark:text-blue-400'
                  : 'border-transparent text-gray-500 hover:text-gray-700 dark:hover:text-
                      gray-300'
              }`}
            &gt;
              ■■■ Owner / Admin
            &lt;/button&gt;
          &lt;/div&gt;

          {/* Title &amp; Mode switch for student */}
          &lt;div className="mb-4"&gt;
            &lt;h3 className="font-bold text-xl text-center"&gt;
              {authRole === 'owner'
                ? 'Owner / Teacher Login'
                : authMode === 'login'
                ? 'Student Login'
                : 'Create Student Account'}
            &lt;/h3&gt;
            {authRole === 'student' &amp;&amp; (
              &lt;p className="text-center text-xs text-gray-500 dark:text-gray-400 mt-1"&gt;
                {authMode === 'login' ? (
```

```
173 |                   &lt;&gt;
174 |                     Don't have an account?{' '}
175 |                     &lt;button
176 |                       type="button"
177 |                       onClick={() =&gt; handleModeChange('register')}
178 |                       className="text-blue-600 hover:underline font-semibold cursor-pointer"
179 |                     &gt;
180 |                       Register here
181 |                     &lt;/button&gt;
182 |                   &lt;/&gt;
183 |                 ) : (
184 |                   &lt;&gt;
185 |                     Already registered?{' '}
186 |                     &lt;button
187 |                       type="button"
188 |                       onClick={() =&gt; handleModeChange('login')}
189 |                       className="text-blue-600 hover:underline font-semibold cursor-pointer"
190 |                     &gt;
191 |                       Login here
192 |                     &lt;/button&gt;
193 |                   &lt;/&gt;
194 |                 )}
195 |               &lt;/p&gt;
196 |             )}
197 |           &lt;/div&gt;
198 |
199 |           {/* Status Message */}
200 |           {serverMsg.text &amp;&amp; (
201 |             &lt;div
202 |               className={`p-3 rounded-lg text-sm mb-4 ${
203 |                 serverMsg.type === 'error'
204 |                   ? 'bg-red-100 text-red-700 border border-red-300 dark:bg-red-900/30
205 |                       dark:text-red-400'
206 |                   : 'bg-green-100 text-green-700 border border-green-300 dark:bg-green-900/30
207 |                       dark:text-green-400'
208 |               }`}
209 |             &gt;
210 |               {serverMsg.text}
211 |             &lt;/div&gt;
212 |           )}
213 |
214 |           &lt;form onSubmit={handleSubmit(onSubmit)} className="space-y-3"&gt;
215 |             {/* Student Registration additional fields */}
216 |             {authRole === 'student' &amp;&amp; authMode === 'register' &amp;&amp; (
217 |               &lt;&gt;
218 |                 &lt;div&gt;
219 |                   &lt;label className="block text-xs font-semibold uppercase tracking-wider mb-1
220 |                       text-gray-600 dark:text-gray-300"&gt;
221 |                     Full Name *
222 |                   &lt;/label&gt;
223 |                   &lt;input
224 |                     {...register('fullname', { required: 'Full name is required' })}
225 |                     type="text"
226 |                     placeholder="Enter your full name"
227 |                     className="w-full px-3 py-2 rounded-lg bg-gray-50 dark:bg-gray-800 border
228 |                         border-gray-300 dark:border-gray-700 text-sm focus:outline-none
229 |                         focus:ring-2 focus:ring-blue-500"
230 |                   /&gt;
231 |                   {errors.fullname &amp;&amp; (
232 |                     &lt;span className="text-xs text-red-500 mt-0.5
233 |                         block"&gt;{errors.fullname.message}&lt;/span&gt;
234 |                   )}
235 |                 &lt;/div&gt;
236 |
237 |                 &lt;div className="grid grid-cols-2 gap-2"&gt;
238 |                   &lt;div&gt;
239 |                     &lt;label className="block text-xs font-semibold uppercase tracking-wider
240 |                         mb-1 text-gray-600 dark:text-gray-300"&gt;
241 |                       Phone Number
242 |                     &lt;/label&gt;
243 |                     &lt;input
244 |                       {...register('phone')}
245 |                       type="tel"
246 |                       placeholder="10-digit number"
247 |                       className="w-full px-3 py-2 rounded-lg bg-gray-50 dark:bg-gray-800
248 |                           border border-gray-300 dark:border-gray-700 text-sm focus:outline-
249 |                           none focus:ring-2 focus:ring-blue-500"
250 |                     /&gt;
251 |                   &lt;/div&gt;
252 |
253 |                   &lt;div&gt;
254 |                     &lt;label className="block text-xs font-semibold uppercase tracking-wider
```

```
                        mb-1 text-gray-600 dark:text-gray-300"&gt;
                      Class / Target
                    &lt;/label&gt;
                    &lt;select
                      {...register('classGrade')}
                      defaultValue="Class 12"
                      className="w-full px-3 py-2 rounded-lg bg-gray-50 dark:bg-gray-800
                          border border-gray-300 dark:border-gray-700 text-sm focus:outline-
                          none focus:ring-2 focus:ring-blue-500"
                    &gt;
                      &lt;option value="Class 11"&gt;Class 11&lt;/option&gt;
                      &lt;option value="Class 12"&gt;Class 12&lt;/option&gt;
                      &lt;option value="JEE Dropper / Target"&gt;JEE Dropper&lt;/option&gt;
                      &lt;option value="NEET"&gt;NEET Target&lt;/option&gt;
                      &lt;option value="Foundation"&gt;Foundation (9/10)&lt;/option&gt;
                    &lt;/select&gt;
                  &lt;/div&gt;
                &lt;/div&gt;
              &lt;/&gt;
            )}

            {/* Email Field */}
            &lt;div&gt;
              &lt;label className="block text-xs font-semibold uppercase tracking-wider mb-1
                  text-gray-600 dark:text-gray-300"&gt;
                Email Address *
              &lt;/label&gt;
              &lt;input
                {...register('email', {
                  required: 'Email is required',
                  pattern: {
                    value: /^\S+@\S+\.\S+$/,
                    message: 'Please enter a valid email address'
                  }
                })}
                type="email"
                placeholder="name@example.com"
                className="w-full px-3 py-2 rounded-lg bg-gray-50 dark:bg-gray-800 border
                    border-gray-300 dark:border-gray-700 text-sm focus:outline-none
                    focus:ring-2 focus:ring-blue-500"
              /&gt;
              {errors.email &amp;&amp; (
                &lt;span className="text-xs text-red-500 mt-0.5
                    block"&gt;{errors.email.message}&lt;/span&gt;
              )}
            &lt;/div&gt;

            {/* Password Field */}
            &lt;div&gt;
              &lt;label className="block text-xs font-semibold uppercase tracking-wider mb-1
                  text-gray-600 dark:text-gray-300"&gt;
                Password *
              &lt;/label&gt;
              &lt;input
                {...register('password', {
                  required: 'Password is required',
                  minLength: {
                    value: 6,
                    message: 'Password must be at least 6 characters'
                  }
                })}
                type="password"
                placeholder="••••••••"
                className="w-full px-3 py-2 rounded-lg bg-gray-50 dark:bg-gray-800 border
                    border-gray-300 dark:border-gray-700 text-sm focus:outline-none
                    focus:ring-2 focus:ring-blue-500"
              /&gt;
              {errors.password &amp;&amp; (
                &lt;span className="text-xs text-red-500 mt-0.5
                    block"&gt;{errors.password.message}&lt;/span&gt;
              )}
            &lt;/div&gt;

            {/* Submit Button */}
            &lt;div className="pt-2"&gt;
              &lt;button
                type="submit"
                disabled={isLoading}
                className="w-full py-2.5 px-4 bg-blue-600 hover:bg-blue-700 disabled:bg-
                    blue-400 text-white font-semibold rounded-lg shadow transition
                    duration-200 cursor-pointer flex justify-center items-center"
              &gt;
```

```
337 |                 {isLoading ? (
338 |                   &lt;span className="inline-block animate-spin mr-2"&gt;■&lt;/span&gt;
339 |                 ) : null}
340 |                 {authRole === 'owner'
341 |                   ? 'Login as Owner'
342 |                   : authMode === 'login'
343 |                   ? 'Login to Student Account'
344 |                   : 'Complete Registration'}
345 |               &lt;/button&gt;
346 |             &lt;/div&gt;
347 |           &lt;/form&gt;
348 |         &lt;/div&gt;
349 |       &lt;/dialog&gt;
350 |     &lt;/div&gt;
351 |   );
352 | }
353 |
354 | export default Login;
```

■ Frontend/src/components/Resources.jsx

Purpose: *Public study materials hub: live search bar by chapter/title, filter chips by subject and class, human-readable file size calculation, and action buttons for inline viewing and downloading.*

```
  1 | import React, { useEffect, useState } from 'react';
  2 | import axios from 'axios';
  3 | import Navbar from './Navbar';
  4 | import Footer from './Footer';
  5 | import { API_BASE_URL } from '../utils/api';
  6 |
  7 | function Resources() {
  8 |   const [resources, setResources] = useState([]);
  9 |   const [loading, setLoading] = useState(true);
 10 |   const [error, setError] = useState('');
 11 |   const [search, setSearch] = useState('');
 12 |   const [selectedSubject, setSelectedSubject] = useState('All');
 13 |   const [selectedClass, setSelectedClass] = useState('All');
 14 |
 15 |   const subjects = [
 16 |     'All',
 17 |     'Physics',
 18 |     'Mechanics',
 19 |     'Electromagnetism',
 20 |     'Optics &amp; Waves',
 21 |     'Thermodynamics',
 22 |     'Modern Physics'
 23 |   ];
 24 |
 25 |   const classes = [
 26 |     'All',
 27 |     'Class 11',
 28 |     'Class 12',
 29 |     'JEE Main &amp; Advanced',
 30 |     'NEET',
 31 |     'All Classes'
 32 |   ];
 33 |
 34 |   const fetchResources = async () =&gt; {
 35 |     setLoading(true);
 36 |     setError('');
 37 |     try {
 38 |       let queryParams = [];
 39 |       if (selectedSubject !== 'All')
 40 |           queryParams.push(`subject=${encodeURIComponent(selectedSubject)}`);
 41 |       if (selectedClass !== 'All')
 42 |           queryParams.push(`classGrade=${encodeURIComponent(selectedClass)}`);
 43 |       if (search.trim() !== '')
 44 |           queryParams.push(`search=${encodeURIComponent(search.trim())}`);
 45 |
 46 |       const queryString = queryParams.length &gt; 0 ? `?${queryParams.join('&amp;')}` : '';
 47 |       const response = await axios.get(`${API_BASE_URL}/api/resources${queryString}`);
 48 |
 49 |       if (response.data &amp;&amp; response.data.success) {
 50 |         setResources(response.data.resources || []);
 51 |       } else {
 52 |         setResources([]);
 53 |       }
 54 |     } catch (err) {
 55 |       console.error('Failed to load resources:', err);
 56 |       setError('Unable to load resources at this moment. Please check back soon.');
```

```
    } finally {
      setLoading(false);
    }
  };

  useEffect(() =&gt; {
    fetchResources();
  }, [selectedSubject, selectedClass]);

  const handleSearchSubmit = (e) =&gt; {
    e.preventDefault();
    fetchResources();
  };

  const formatFileSize = (bytes) =&gt; {
    if (!bytes || bytes === 0) return 'Unknown size';
    const k = 1024;
    const sizes = ['Bytes', 'KB', 'MB', 'GB'];
    const i = Math.floor(Math.log(bytes) / Math.log(k));
    return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + ' ' + sizes[i];
  };

  const formatDate = (dateString) =&gt; {
    if (!dateString) return '';
    const date = new Date(dateString);
    return date.toLocaleDateString('en-IN', {
      day: 'numeric',
      month: 'short',
      year: 'numeric'
    });
  };

  return (
    &lt;div className="min-h-screen flex flex-col bg-gray-50 dark:bg-gray-900 text-gray-800
        dark:text-gray-100"&gt;
      &lt;Navbar /&gt;

      &lt;main className="flex-grow pt-24 pb-16 px-4 md:px-12 max-w-7xl mx-auto w-full"&gt;
        {/* Hero Header */}
        &lt;div className="text-center max-w-3xl mx-auto mb-10"&gt;
          &lt;div className="inline-block bg-blue-100 dark:bg-blue-900/40 text-blue-700
              dark:text-blue-300 text-xs font-semibold uppercase tracking-wider px-3 py-1
              rounded-full mb-3"&gt;
            Official Course Materials
          &lt;/div&gt;
          &lt;h1 className="text-3xl md:text-5xl font-extrabold tracking-tight text-gray-900
              dark:text-white"&gt;
            Study Resources &amp;amp; Lecture Notes
          &lt;/h1&gt;
          &lt;p className="mt-3 text-base md:text-lg text-gray-600 dark:text-gray-300"&gt;
            Download PDFs of lecture notes, HCV &amp;amp; Irodov practice sheets, revision mindmaps,
               and formula sheets curated by JD Sir.
          &lt;/p&gt;
        &lt;/div&gt;

        {/* Search &amp;amp; Filter Bar */}
        &lt;div className="bg-white dark:bg-gray-800 rounded-2xl shadow-sm border border-gray-200
            dark:border-gray-700 p-4 md:p-6 mb-8"&gt;
          &lt;form onSubmit={handleSearchSubmit} className="flex flex-col md:flex-row gap-3
              mb-4"&gt;
            &lt;div className="relative flex-grow"&gt;
              &lt;span className="absolute inset-y-0 left-0 flex items-center pl-3 text-
                  gray-400"&gt;
                ■
              &lt;/span&gt;
              &lt;input
                type="text"
                placeholder="Search notes by title, topic, or chapter..."
                value={search}
                onChange={(e) =&gt; setSearch(e.target.value)}
                className="w-full pl-10 pr-4 py-2.5 rounded-xl bg-gray-50 dark:bg-gray-900
                    border border-gray-300 dark:border-gray-600 focus:outline-none
                    focus:ring-2 focus:ring-blue-500 text-sm"
              /&gt;
            &lt;/div&gt;
            &lt;button
              type="submit"
              className="px-6 py-2.5 bg-blue-600 hover:bg-blue-700 text-white font-medium
                  rounded-xl text-sm transition-colors shadow cursor-pointer"
            &gt;
              Search
            &lt;/button&gt;
```

```
            {(search || selectedSubject !== 'All' || selectedClass !== 'All') &amp;&amp; (
              &lt;button
                type="button"
                onClick={() =&gt; {
                  setSearch('');
                  setSelectedSubject('All');
                  setSelectedClass('All');
                }}
                className="px-4 py-2.5 bg-gray-200 dark:bg-gray-700 hover:bg-gray-300
                    dark:hover:bg-gray-600 text-gray-700 dark:text-gray-200 font-medium
                    rounded-xl text-sm transition-colors cursor-pointer"
              &gt;
                Reset
              &lt;/button&gt;
            )}
          &lt;/form&gt;

          {/* Filter Chips */}
          &lt;div className="space-y-3"&gt;
            {/* Subject Filters */}
            &lt;div className="flex flex-wrap items-center gap-2"&gt;
              &lt;span className="text-xs font-semibold uppercase tracking-wider text-gray-500
                  dark:text-gray-400 mr-1"&gt;
                Subject:
              &lt;/span&gt;
              {subjects.map((sub) =&gt; (
                &lt;button
                  key={sub}
                  type="button"
                  onClick={() =&gt; setSelectedSubject(sub)}
                  className={`text-xs px-3 py-1.5 rounded-full transition-all cursor-pointer
                      font-medium ${
                    selectedSubject === sub
                      ? 'bg-blue-600 text-white shadow-sm'
                      : 'bg-gray-100 dark:bg-gray-700 text-gray-600 dark:text-gray-300
                          hover:bg-gray-200 dark:hover:bg-gray-600'
                  }`}
                &gt;
                  {sub}
                &lt;/button&gt;
              ))}
            &lt;/div&gt;

            {/* Class/Target Filters */}
            &lt;div className="flex flex-wrap items-center gap-2"&gt;
              &lt;span className="text-xs font-semibold uppercase tracking-wider text-gray-500
                  dark:text-gray-400 mr-1"&gt;
                Target:
              &lt;/span&gt;
              {classes.map((cls) =&gt; (
                &lt;button
                  key={cls}
                  type="button"
                  onClick={() =&gt; setSelectedClass(cls)}
                  className={`text-xs px-3 py-1.5 rounded-full transition-all cursor-pointer
                      font-medium ${
                    selectedClass === cls
                      ? 'bg-indigo-600 text-white shadow-sm'
                      : 'bg-gray-100 dark:bg-gray-700 text-gray-600 dark:text-gray-300
                          hover:bg-gray-200 dark:hover:bg-gray-600'
                  }`}
                &gt;
                  {cls}
                &lt;/button&gt;
              ))}
            &lt;/div&gt;
          &lt;/div&gt;
        &lt;/div&gt;

        {/* Resources Grid */}
        {loading ? (
          &lt;div className="flex flex-col items-center justify-center py-20"&gt;
            &lt;span className="loading loading-spinner loading-lg text-blue-600"&gt;&lt;/span&gt;
            &lt;p className="mt-4 text-sm text-gray-500"&gt;Loading study materials...&lt;/p&gt;
          &lt;/div&gt;
        ) : error ? (
          &lt;div className="bg-red-50 dark:bg-red-900/20 border border-red-200 dark:border-
              red-800 text-red-600 dark:text-red-300 p-6 rounded-2xl text-center my-6"&gt;
            &lt;p className="font-semibold"&gt;{error}&lt;/p&gt;
            &lt;button
              onClick={fetchResources}
              className="mt-3 px-4 py-2 bg-red-600 hover:bg-red-700 text-white rounded-lg
```

```
                  text-xs font-medium"
              &gt;
                Retry
              &lt;/button&gt;
            &lt;/div&gt;
          ) : resources.length === 0 ? (
            &lt;div className="bg-white dark:bg-gray-800 rounded-2xl border border-gray-200
                dark:border-gray-700 p-12 text-center my-6"&gt;
              &lt;div className="text-5xl mb-3"&gt;■&lt;/div&gt;
              &lt;h3 className="text-lg font-bold text-gray-800 dark:text-gray-200"&gt;
                No Study Materials Found
              &lt;/h3&gt;
              &lt;p className="text-sm text-gray-500 dark:text-gray-400 max-w-md mx-auto mt-2"&gt;
                There are currently no uploaded PDF notes matching your filter criteria. Try
                    resetting filters or check back later!
              &lt;/p&gt;
            &lt;/div&gt;
          ) : (
            &lt;div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6"&gt;
              {resources.map((item) =&gt; (
                &lt;div
                  key={item._id}
                  className="bg-white dark:bg-gray-800 rounded-2xl border border-gray-200
                      dark:border-gray-700 shadow-sm hover:shadow-md transition-shadow flex
                      flex-col justify-between overflow-hidden"
                &gt;
                  &lt;div className="p-6"&gt;
                    {/* Badges */}
                    &lt;div className="flex flex-wrap items-center gap-2 mb-3"&gt;
                      &lt;span className="bg-blue-100 dark:bg-blue-900/40 text-blue-700 dark:text-
                          blue-300 text-xs font-semibold px-2.5 py-1 rounded-md"&gt;
                        {item.subject || 'Physics'}
                      &lt;/span&gt;
                      &lt;span className="bg-purple-100 dark:bg-purple-900/40 text-purple-700
                          dark:text-purple-300 text-xs font-semibold px-2.5 py-1 rounded-md"&gt;
                        {item.classGrade || 'All Classes'}
                      &lt;/span&gt;
                    &lt;/div&gt;

                    {/* Title &amp; Chapter */}
                    &lt;h3 className="text-lg font-bold text-gray-900 dark:text-white leading-
                        snug"&gt;
                      {item.title}
                    &lt;/h3&gt;
                    {item.chapter &amp;&amp; (
                      &lt;p className="text-xs font-medium text-blue-600 dark:text-blue-400 mt-1"&gt;
                        ■ Chapter: {item.chapter}
                      &lt;/p&gt;
                    )}

                    {/* Description */}
                    {item.description &amp;&amp; (
                      &lt;p className="text-sm text-gray-600 dark:text-gray-300 mt-2 line-clamp-3"&gt;
                        {item.description}
                      &lt;/p&gt;
                    )}

                    {/* File Metadata */}
                    &lt;div className="mt-4 pt-3 border-t border-gray-100 dark:border-gray-700/60
                        flex items-center justify-between text-xs text-gray-500 dark:text-
                        gray-400"&gt;
                      &lt;span className="flex items-center gap-1"&gt;
                        ■ {formatFileSize(item.pdf?.fileSize)}
                      &lt;/span&gt;
                      &lt;span&gt;{formatDate(item.createdAt)}&lt;/span&gt;
                    &lt;/div&gt;
                  &lt;/div&gt;

                  {/* Card Action Buttons */}
                  &lt;div className="bg-gray-50 dark:bg-gray-850 p-4 border-t border-gray-100
                      dark:border-gray-700 grid grid-cols-2 gap-2"&gt;
                    &lt;a
                      href={`${API_BASE_URL}/api/resources/${item._id}/view`}
                      target="_blank"
                      rel="noopener noreferrer"
                      className="py-2 px-3 text-center bg-white dark:bg-gray-700 hover:bg-
                          gray-100 dark:hover:bg-gray-600 text-blue-600 dark:text-blue-300
                          border border-gray-300 dark:border-gray-600 rounded-xl text-xs font-
                          semibold transition duration-150 flex items-center justify-center
                          gap-1.5"
                    &gt;
                      &lt;span&gt;■■&lt;/span&gt; View PDF
```

```
303 |                   &lt;/a&gt;
304 |                   &lt;a
305 |                     href={`${API_BASE_URL}/api/resources/${item._id}/download?download=true`}
306 |                     download
307 |                     className="py-2 px-3 text-center bg-blue-600 hover:bg-blue-700 text-white
308 |                         rounded-xl text-xs font-semibold transition duration-150 flex items-
309 |                         center justify-center gap-1.5 shadow"
310 |                   &gt;
311 |                     &lt;span&gt;■■&lt;/span&gt; Download
312 |                   &lt;/a&gt;
313 |                 &lt;/div&gt;
314 |               &lt;/div&gt;
315 |             ))}
316 |           &lt;/div&gt;
317 |         )}
318 |       &lt;/main&gt;
319 |
320 |       &lt;Footer /&gt;
321 |     &lt;/div&gt;
322 |   );
323 | }
324 |
325 | export default Resources;
```

■ Frontend/src/components/OwnerResourceUpload.jsx

Purpose: *Admin PDF study material manager: drag-and-drop 30MB PDF file picker with form inputs, upload progress state, and live table of uploaded materials with delete buttons.*

```
  1 | import React, { useState, useEffect } from 'react';
  2 | import axios from 'axios';
  3 | import { useForm } from 'react-hook-form';
  4 | import { Link } from 'react-router-dom';
  5 | import { API_BASE_URL } from '../utils/api';
  6 |
  7 | function OwnerResourceUpload() {
  8 |   const [resources, setResources] = useState([]);
  9 |   const [isUploading, setIsUploading] = useState(false);
 10 |   const [selectedFile, setSelectedFile] = useState(null);
 11 |   const [statusMsg, setStatusMsg] = useState({ text: '', type: '' });
 12 |
 13 |   const {
 14 |     register,
 15 |     handleSubmit,
 16 |     reset,
 17 |     formState: { errors }
 18 |   } = useForm();
 19 |
 20 |   const fetchResources = async () =&gt; {
 21 |     try {
 22 |       const res = await axios.get(`${API_BASE_URL}/api/resources`);
 23 |       if (res.data &amp;&amp; res.data.success) {
 24 |         setResources(res.data.resources || []);
 25 |       }
 26 |     } catch (err) {
 27 |       console.error('Error fetching resources:', err);
 28 |     }
 29 |   };
 30 |
 31 |   useEffect(() =&gt; {
 32 |     fetchResources();
 33 |   }, []);
 34 |
 35 |   const handleFileChange = (e) =&gt; {
 36 |     if (e.target.files &amp;&amp; e.target.files[0]) {
 37 |       const file = e.target.files[0];
 38 |       if (file.type !== 'application/pdf' &amp;&amp; !file.name.toLowerCase().endsWith('.pdf')) {
 39 |         alert('Please choose a valid PDF file (.pdf)');
 40 |         e.target.value = null;
 41 |         setSelectedFile(null);
 42 |         return;
 43 |       }
 44 |       setSelectedFile(file);
 45 |     }
 46 |   };
 47 |
 48 |   const onSubmit = async (data) =&gt; {
 49 |     if (!selectedFile) {
 50 |       setStatusMsg({ text: 'Please select a PDF file to upload', type: 'error' });
 51 |       return;
```

```
    }

    setIsUploading(true);
    setStatusMsg({ text: '', type: '' });

    const formData = new FormData();
    formData.append('title', data.title);
    formData.append('subject', data.subject);
    formData.append('classGrade', data.classGrade);
    formData.append('chapter', data.chapter || '');
    formData.append('description', data.description || '');
    formData.append('pdf', selectedFile);

    try {
      const res = await axios.post(`${API_BASE_URL}/api/resources/upload`, formData, {
        headers: { 'Content-Type': 'multipart/form-data' }
      });

      if (res.data &amp;&amp; res.data.success) {
        setStatusMsg({ text: 'PDF Class Material uploaded successfully!', type: 'success' });
        reset();
        setSelectedFile(null);
        // Clear file input manually
        const fileInput = document.getElementById('pdf-file-input');
        if (fileInput) fileInput.value = null;
        fetchResources();
      } else {
        setStatusMsg({ text: res.data.message || 'Upload failed', type: 'error' });
      }
    } catch (err) {
      console.error('Upload error:', err);
      setStatusMsg({
        text: err.response?.data?.message || 'Upload failed. Please check the file and try
            again.',
        type: 'error'
      });
    } finally {
      setIsUploading(false);
    }
  };

  const handleDelete = async (id, title) =&gt; {
    if (!window.confirm(`Are you sure you want to delete "${title}"?`)) {
      return;
    }

    try {
      const res = await axios.delete(`${API_BASE_URL}/api/resources/${id}`);
      if (res.data &amp;&amp; res.data.success) {
        setStatusMsg({ text: `"${title}" was deleted successfully.`, type: 'success' });
        fetchResources();
      }
    } catch (err) {
      console.error('Delete error:', err);
      alert('Failed to delete resource');
    }
  };

  const formatFileSize = (bytes) =&gt; {
    if (!bytes || bytes === 0) return '0 KB';
    const k = 1024;
    const sizes = ['Bytes', 'KB', 'MB', 'GB'];
    const i = Math.floor(Math.log(bytes) / Math.log(k));
    return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + ' ' + sizes[i];
  };

  return (
    &lt;div className="min-h-screen bg-gray-900 text-gray-100 p-4 md:p-8"&gt;
      {/* Header bar */}
      &lt;div className="max-w-5xl mx-auto flex flex-col sm:flex-row justify-between items-center
          mb-8 gap-4 border-b border-gray-800 pb-4"&gt;
        &lt;div&gt;
          &lt;h1 className="text-2xl md:text-3xl font-bold text-white flex items-center gap-2"&gt;
            &lt;span&gt;■&lt;/span&gt; Admin - Upload Class Material (PDF)
          &lt;/h1&gt;
          &lt;p className="text-sm text-gray-400 mt-1"&gt;
            Upload notes, lecture slides, and question sheets for students to download.
          &lt;/p&gt;
        &lt;/div&gt;
        &lt;div className="flex gap-2"&gt;
          &lt;Link
            to="/owner"
```

```
134 |             className="px-4 py-2 bg-gray-800 hover:bg-gray-700 text-gray-300 rounded-lg text-
135 |                 sm font-medium transition"
136 |           &gt;
137 |             ← Back to Dashboard
138 |           &lt;/Link&gt;
139 |           &lt;Link
140 |             to="/resources"
141 |             className="px-4 py-2 bg-blue-600 hover:bg-blue-700 text-white rounded-lg text-sm
142 |                 font-medium transition"
143 |           &gt;
144 |             View Public Resources
145 |           &lt;/Link&gt;
146 |         &lt;/div&gt;
147 |       &lt;/div&gt;
148 |
149 |       &lt;div className="max-w-5xl mx-auto grid grid-cols-1 lg:grid-cols-12 gap-8"&gt;
150 |         {/* Upload Form (Left Column) */}
151 |         &lt;div className="lg:col-span-5"&gt;
152 |           &lt;div className="bg-gray-800 border border-gray-700 rounded-2xl p-6 shadow-xl sticky
153 |               top-8"&gt;
154 |             &lt;h2 className="text-xl font-bold mb-4 text-white flex items-center gap-2"&gt;
155 |               &lt;span&gt;■&lt;/span&gt; Upload New PDF Material
156 |             &lt;/h2&gt;
157 |
158 |             {statusMsg.text &amp;&amp; (
159 |               &lt;div
160 |                 className={`p-3 rounded-xl text-sm mb-4 ${
161 |                   statusMsg.type === 'error'
162 |                     ? 'bg-red-900/40 border border-red-700 text-red-300'
163 |                     : 'bg-green-900/40 border border-green-700 text-green-300'
164 |                 }`}
165 |               &gt;
166 |                 {statusMsg.text}
167 |               &lt;/div&gt;
168 |             )}
169 |
170 |             &lt;form onSubmit={handleSubmit(onSubmit)} className="space-y-4"&gt;
171 |               &lt;div&gt;
172 |                 &lt;label className="block text-xs font-semibold uppercase tracking-wider text-
173 |                     gray-300 mb-1"&gt;
174 |                   Material Title *
175 |                 &lt;/label&gt;
176 |                 &lt;input
177 |                   {...register('title', { required: 'Title is required' })}
178 |                   placeholder="e.g. Kinematics Lecture Notes &amp; Problem Sheet"
179 |                   className="w-full px-3.5 py-2.5 bg-gray-900 border border-gray-700 rounded-
180 |                       xl text-sm text-white focus:outline-none focus:ring-2 focus:ring-
181 |                       blue-500"
182 |                 /&gt;
183 |                 {errors.title &amp;&amp; (
184 |                   &lt;span className="text-xs text-red-400 mt-1
185 |                       block"&gt;{errors.title.message}&lt;/span&gt;
186 |                 )}
187 |               &lt;/div&gt;
188 |
189 |               &lt;div className="grid grid-cols-2 gap-3"&gt;
190 |                 &lt;div&gt;
191 |                   &lt;label className="block text-xs font-semibold uppercase tracking-wider text-
192 |                       gray-300 mb-1"&gt;
193 |                     Subject *
194 |                   &lt;/label&gt;
195 |                   &lt;select
196 |                     {...register('subject')}
197 |                     defaultValue="Physics"
198 |                     className="w-full px-3 py-2.5 bg-gray-900 border border-gray-700 rounded-
199 |                         xl text-sm text-white focus:outline-none focus:ring-2 focus:ring-
200 |                         blue-500"
201 |                   &gt;
202 |                     &lt;option value="Physics"&gt;Physics&lt;/option&gt;
203 |                     &lt;option value="Mechanics"&gt;Mechanics&lt;/option&gt;
204 |                     &lt;option value="Electromagnetism"&gt;Electromagnetism&lt;/option&gt;
205 |                     &lt;option value="Optics &amp; Waves"&gt;Optics &amp; Waves&lt;/option&gt;
206 |                     &lt;option value="Thermodynamics"&gt;Thermodynamics&lt;/option&gt;
207 |                     &lt;option value="Modern Physics"&gt;Modern Physics&lt;/option&gt;
208 |                   &lt;/select&gt;
209 |                 &lt;/div&gt;
210 |
211 |                 &lt;div&gt;
212 |                   &lt;label className="block text-xs font-semibold uppercase tracking-wider text-
213 |                       gray-300 mb-1"&gt;
214 |                     Target Class *
215 |                   &lt;/label&gt;
```

```
                  &lt;select
                    {...register('classGrade')}
                    defaultValue="Class 12"
                    className="w-full px-3 py-2.5 bg-gray-900 border border-gray-700 rounded-
                        xl text-sm text-white focus:outline-none focus:ring-2 focus:ring-
                        blue-500"
                  &gt;
                    &lt;option value="All Classes"&gt;All Classes&lt;/option&gt;
                    &lt;option value="Class 11"&gt;Class 11&lt;/option&gt;
                    &lt;option value="Class 12"&gt;Class 12&lt;/option&gt;
                    &lt;option value="JEE Main &amp; Advanced"&gt;JEE Main &amp; Adv&lt;/option&gt;
                    &lt;option value="NEET"&gt;NEET Target&lt;/option&gt;
                  &lt;/select&gt;
                &lt;/div&gt;
              &lt;/div&gt;

              &lt;div&gt;
                &lt;label className="block text-xs font-semibold uppercase tracking-wider text-
                    gray-300 mb-1"&gt;
                  Chapter / Topic (Optional)
                &lt;/label&gt;
                &lt;input
                  {...register('chapter')}
                  placeholder="e.g. Work, Energy and Power"
                  className="w-full px-3.5 py-2.5 bg-gray-900 border border-gray-700 rounded-
                      xl text-sm text-white focus:outline-none focus:ring-2 focus:ring-
                      blue-500"
                /&gt;
              &lt;/div&gt;

              &lt;div&gt;
                &lt;label className="block text-xs font-semibold uppercase tracking-wider text-
                    gray-300 mb-1"&gt;
                  Description / Hints (Optional)
                &lt;/label&gt;
                &lt;textarea
                  {...register('description')}
                  rows="2"
                  placeholder="e.g. Includes Irodov problems 1.1 to 1.35 solved with steps"
                  className="w-full px-3.5 py-2.5 bg-gray-900 border border-gray-700 rounded-
                      xl text-sm text-white focus:outline-none focus:ring-2 focus:ring-
                      blue-500"
                &gt;&lt;/textarea&gt;
              &lt;/div&gt;

              {/* PDF File Picker */}
              &lt;div&gt;
                &lt;label className="block text-xs font-semibold uppercase tracking-wider text-
                    gray-300 mb-1"&gt;
                  Select PDF File * (Max 30 MB)
                &lt;/label&gt;
                &lt;div className="border-2 border-dashed border-gray-600 rounded-xl p-4 text-
                    center hover:border-blue-500 transition-colors bg-gray-900/50"&gt;
                  &lt;input
                    id="pdf-file-input"
                    type="file"
                    accept=".pdf,application/pdf"
                    onChange={handleFileChange}
                    className="block w-full text-xs text-gray-400 file:mr-3 file:py-2
                        file:px-4 file:rounded-lg file:border-0 file:text-xs file:font-
                        semibold file:bg-blue-600 file:text-white hover:file:bg-blue-700
                        cursor-pointer"
                  /&gt;
                  {selectedFile &amp;&amp; (
                    &lt;div className="mt-2 text-xs text-green-400 flex items-center justify-
                        center gap-1"&gt;
                      &lt;span&gt;✓&lt;/span&gt; Selected: {selectedFile.name}
                          ({formatFileSize(selectedFile.size)})
                    &lt;/div&gt;
                  )}
                &lt;/div&gt;
              &lt;/div&gt;

              &lt;button
                type="submit"
                disabled={isUploading}
                className="w-full py-3 bg-blue-600 hover:bg-blue-700 disabled:bg-blue-800
                    text-white font-semibold rounded-xl transition duration-150 shadow-lg
                    cursor-pointer flex justify-center items-center gap-2"
              &gt;
                {isUploading ? (
                  &lt;&gt;
```

```
298 |                     &lt;span className="loading loading-spinner loading-sm"&gt;&lt;/span&gt;
299 |                     &lt;span&gt;Uploading PDF to Database...&lt;/span&gt;
300 |                   &lt;/&gt;
301 |                 ) : (
302 |                   &lt;&gt;
303 |                     &lt;span&gt;■&lt;/span&gt;
304 |                     &lt;span&gt;Upload Resource&lt;/span&gt;
305 |                   &lt;/&gt;
306 |                 )}
307 |               &lt;/button&gt;
308 |             &lt;/form&gt;
309 |           &lt;/div&gt;
310 |         &lt;/div&gt;
311 |
312 |         {/* Existing Uploaded Resources (Right Column) */}
313 |         &lt;div className="lg:col-span-7"&gt;
314 |           &lt;div className="bg-gray-800 border border-gray-700 rounded-2xl p-6 shadow-xl"&gt;
315 |             &lt;div className="flex justify-between items-center mb-6"&gt;
316 |               &lt;h2 className="text-xl font-bold text-white flex items-center gap-2"&gt;
317 |                 &lt;span&gt;■&lt;/span&gt; Uploaded Materials ({resources.length})
318 |               &lt;/h2&gt;
319 |               &lt;button
320 |                 onClick={fetchResources}
321 |                 className="text-xs text-gray-400 hover:text-white flex items-center gap-1
322 |                     cursor-pointer bg-gray-700 px-3 py-1.5 rounded-lg"
323 |               &gt;
324 |                 &lt;span&gt;■&lt;/span&gt; Refresh
325 |               &lt;/button&gt;
326 |             &lt;/div&gt;
327 |
328 |             {resources.length === 0 ? (
329 |               &lt;div className="py-12 text-center text-gray-400"&gt;
330 |                 &lt;p className="text-3xl mb-2"&gt;■&lt;/p&gt;
331 |                 &lt;p&gt;No materials uploaded yet.&lt;/p&gt;
332 |                 &lt;p className="text-xs text-gray-500 mt-1"&gt;Use the form on the left to upload
333 |                     your first class PDF!&lt;/p&gt;
334 |               &lt;/div&gt;
335 |             ) : (
336 |               &lt;div className="space-y-4"&gt;
337 |                 {resources.map((item) =&gt; (
338 |                   &lt;div
339 |                     key={item._id}
340 |                     className="p-4 bg-gray-900 rounded-xl border border-gray-700/80 flex flex-
341 |                         col sm:flex-row justify-between items-start sm:items-center gap-3
342 |                         hover:border-gray-600 transition"
343 |                   &gt;
344 |                     &lt;div className="flex-1"&gt;
345 |                       &lt;div className="flex flex-wrap items-center gap-2 mb-1"&gt;
346 |                         &lt;span className="text-[10px] uppercase font-bold tracking-wider px-2
347 |                             py-0.5 rounded bg-blue-900/60 text-blue-300"&gt;
348 |                           {item.subject}
349 |                         &lt;/span&gt;
350 |                         &lt;span className="text-[10px] uppercase font-bold tracking-wider px-2
351 |                             py-0.5 rounded bg-purple-900/60 text-purple-300"&gt;
352 |                           {item.classGrade}
353 |                         &lt;/span&gt;
354 |                         {item.chapter &amp;&amp; (
355 |                           &lt;span className="text-xs text-gray-400"&gt;
356 |                             • {item.chapter}
357 |                           &lt;/span&gt;
358 |                         )}
359 |                       &lt;/div&gt;
360 |                       &lt;h3 className="font-semibold text-white text-sm"&gt;{item.title}&lt;/h3&gt;
361 |                       &lt;p className="text-xs text-gray-400 mt-1"&gt;
362 |                         ■ {item.pdf?.fileName || 'Document.pdf'}
363 |                           ({formatFileSize(item.pdf?.fileSize)})
364 |                       &lt;/p&gt;
365 |                     &lt;/div&gt;
366 |
367 |                     &lt;div className="flex items-center gap-2 w-full sm:w-auto justify-end"&gt;
368 |                       &lt;a
369 |                         href={`${API_BASE_URL}/api/resources/${item._id}/view`}
370 |                         target="_blank"
371 |                         rel="noopener noreferrer"
372 |                         className="px-3 py-1.5 bg-gray-700 hover:bg-gray-600 text-blue-300
373 |                             rounded-lg text-xs font-medium transition"
374 |                       &gt;
375 |                         ■■ View
376 |                       &lt;/a&gt;
377 |                       &lt;button
378 |                         onClick={() =&gt; handleDelete(item._id, item.title)}
379 |                         className="px-3 py-1.5 bg-red-900/40 hover:bg-red-800 text-red-300
```

```
                            border border-red-800 rounded-lg text-xs font-medium transition
                            cursor-pointer"
                        &gt;
                          ■■ Delete
                        &lt;/button&gt;
                      &lt;/div&gt;
                    &lt;/div&gt;
                  ))}
                &lt;/div&gt;
              )}
            &lt;/div&gt;
          &lt;/div&gt;
        &lt;/div&gt;
      &lt;/div&gt;
    );
}

export default OwnerResourceUpload;
```

■ Frontend/src/components/ViewStudents.jsx

Purpose: *Admin registered students viewer: searchable tabular list displaying all student accounts registered through the portal.*

```
import React, { useEffect, useState } from 'react';
import axios from 'axios';
import { Link } from 'react-router-dom';
import { API_BASE_URL } from '../utils/api';

function ViewStudents() {
  const [students, setStudents] = useState([]);
  const [loading, setLoading] = useState(true);
  const [search, setSearch] = useState('');

  const fetchStudents = async () =&gt; {
    setLoading(true);
    try {
      const res = await axios.get(`${API_BASE_URL}/api/students/all`);
      if (res.data &amp;&amp; res.data.success) {
        setStudents(res.data.students || []);
      }
    } catch (err) {
      console.error('Failed to fetch students:', err);
    } finally {
      setLoading(false);
    }
  };

  useEffect(() =&gt; {
    fetchStudents();
  }, []);

  const filteredStudents = students.filter(
    (s) =&gt;
      s.fullname?.toLowerCase().includes(search.toLowerCase()) ||
      s.email?.toLowerCase().includes(search.toLowerCase()) ||
      s.phone?.includes(search) ||
      s.classGrade?.toLowerCase().includes(search.toLowerCase())
  );

  return (
    &lt;div className="min-h-screen bg-gray-900 text-gray-100 p-4 md:p-8"&gt;
      &lt;div className="max-w-6xl mx-auto"&gt;
        {/* Header */}
        &lt;div className="flex flex-col sm:flex-row justify-between items-center mb-8 gap-4
            border-b border-gray-800 pb-4"&gt;
          &lt;div&gt;
            &lt;h1 className="text-2xl md:text-3xl font-bold text-white flex items-center gap-2"&gt;
              &lt;span&gt;■&lt;/span&gt; Registered Students Database ({students.length})
            &lt;/h1&gt;
            &lt;p className="text-sm text-gray-400 mt-1"&gt;
              List of all student accounts registered through the portal.
            &lt;/p&gt;
          &lt;/div&gt;
          &lt;div className="flex gap-2"&gt;
            &lt;Link
              to="/owner"
              className="px-4 py-2 bg-gray-800 hover:bg-gray-700 text-gray-300 rounded-lg
                  text-sm font-medium transition"
            &gt;
              ← Back to Dashboard
            &lt;/Link&gt;
```

```
          &lt;/div&gt;
        &lt;/div&gt;

        {/* Search */}
        &lt;div className="mb-6 max-w-md"&gt;
          &lt;input
            type="text"
            placeholder="Search by student name, email, phone, or class..."
            value={search}
            onChange={(e) =&gt; setSearch(e.target.value)}
            className="w-full px-4 py-2.5 bg-gray-800 border border-gray-700 rounded-xl text-
                sm text-white focus:outline-none focus:ring-2 focus:ring-blue-500"
          /&gt;
        &lt;/div&gt;

        {/* Student Table */}
        {loading ? (
          &lt;div className="py-20 text-center"&gt;
            &lt;span className="loading loading-spinner loading-lg text-blue-500"&gt;&lt;/span&gt;
            &lt;p className="mt-3 text-sm text-gray-400"&gt;Loading student accounts...&lt;/p&gt;
          &lt;/div&gt;
        ) : filteredStudents.length === 0 ? (
          &lt;div className="bg-gray-800 border border-gray-700 rounded-2xl p-12 text-center
              text-gray-400"&gt;
            &lt;p className="text-4xl mb-2"&gt;■■■&lt;/p&gt;
            &lt;p className="text-lg font-semibold text-white"&gt;No registered students found&lt;/p&gt;
            &lt;p className="text-xs text-gray-400 mt-1"&gt;
              Students will appear here once they register on the website portal.
            &lt;/p&gt;
          &lt;/div&gt;
        ) : (
          &lt;div className="bg-gray-800 border border-gray-700 rounded-2xl overflow-hidden
              shadow-xl"&gt;
            &lt;div className="overflow-x-auto"&gt;
              &lt;table className="w-full text-left text-sm"&gt;
                &lt;thead className="bg-gray-950 text-gray-400 uppercase text-xs tracking-wider"&gt;
                  &lt;tr&gt;
                    &lt;th className="py-3 px-4"&gt;#&lt;/th&gt;
                    &lt;th className="py-3 px-4"&gt;Full Name&lt;/th&gt;
                    &lt;th className="py-3 px-4"&gt;Email&lt;/th&gt;
                    &lt;th className="py-3 px-4"&gt;Phone&lt;/th&gt;
                    &lt;th className="py-3 px-4"&gt;Class / Target&lt;/th&gt;
                    &lt;th className="py-3 px-4"&gt;Registered Date&lt;/th&gt;
                  &lt;/tr&gt;
                &lt;/thead&gt;
                &lt;tbody className="divide-y divide-gray-700"&gt;
                  {filteredStudents.map((s, index) =&gt; (
                    &lt;tr key={s._id} className="hover:bg-gray-750 transition-colors"&gt;
                      &lt;td className="py-3 px-4 text-gray-500"&gt;{index + 1}&lt;/td&gt;
                      &lt;td className="py-3 px-4 font-semibold text-white"&gt;{s.fullname}&lt;/td&gt;
                      &lt;td className="py-3 px-4 text-gray-300"&gt;{s.email}&lt;/td&gt;
                      &lt;td className="py-3 px-4 text-gray-300"&gt;{s.phone || '—'}&lt;/td&gt;
                      &lt;td className="py-3 px-4"&gt;
                        &lt;span className="bg-blue-900/60 text-blue-300 text-xs px-2.5 py-1
                            rounded-md font-medium"&gt;
                          {s.classGrade || 'Class 12'}
                        &lt;/span&gt;
                      &lt;/td&gt;
                      &lt;td className="py-3 px-4 text-gray-400 text-xs"&gt;
                        {s.createdAt ? new Date(s.createdAt).toLocaleDateString('en-IN') :
                            '—'}
                      &lt;/td&gt;
                    &lt;/tr&gt;
                  ))}
                &lt;/tbody&gt;
              &lt;/table&gt;
            &lt;/div&gt;
          &lt;/div&gt;
        )}
      &lt;/div&gt;
    &lt;/div&gt;
  );
}

export default ViewStudents;
```

■ Frontend/src/components/owner.jsx

Purpose: *Admin portal control center: dashboard for JD Sir with quick-navigation cards for PDF uploads, student database, admission applications, and gallery photos.*

```
import React from 'react';
import { Link } from 'react-router-dom';

function Owner() {
  return (
    &lt;div className="min-h-screen bg-gray-900 text-gray-100 py-10 px-4"&gt;
      &lt;div className="max-w-4xl mx-auto"&gt;
        {/* Top Header */}
        &lt;div className="flex justify-between items-center mb-8 border-b border-gray-800 pb-4"&gt;
          &lt;div&gt;
            &lt;h1 className="text-3xl font-extrabold text-white"&gt;JD Sir's Admin Portal&lt;/h1&gt;
            &lt;p className="text-sm text-gray-400 mt-1"&gt;Manage website materials, enrollments,
                and student accounts.&lt;/p&gt;
          &lt;/div&gt;
          &lt;Link
            to="/"
            className="px-4 py-2 bg-gray-800 hover:bg-gray-700 text-gray-300 rounded-lg text-
                sm font-medium transition"
          &gt;
            ← View Main Website
          &lt;/Link&gt;
        &lt;/div&gt;

        {/* Profile Card */}
        &lt;div className="flex flex-col sm:flex-row items-center gap-6 bg-gray-800 border
            border-gray-700 rounded-2xl p-6 mb-10 shadow-xl"&gt;
          &lt;img
            src="/jd.png"
            alt="Owner"
            className="rounded-full shadow-lg w-28 h-28 object-cover border-4 border-blue-500"
          /&gt;
          &lt;div&gt;
            &lt;h2 className="text-2xl font-bold text-white"&gt;Er. Jagannath Dutta (JD Sir)&lt;/h2&gt;
            &lt;p className="text-blue-400 font-medium text-sm"&gt;Physics Mentor | Irodov +
                Classes&lt;/p&gt;
            &lt;p className="text-gray-400 text-xs mt-1"&gt;
              Baripada, Odisha | Preparing future IITians &amp;amp; Doctors
            &lt;/p&gt;
          &lt;/div&gt;
        &lt;/div&gt;

        {/* Action Grid */}
        &lt;div className="grid grid-cols-1 md:grid-cols-2 gap-6"&gt;
          {/* Card 1: Upload Class Material PDF */}
          &lt;Link
            to="/owner/resources"
            className="bg-gray-800 hover:bg-gray-750 border border-blue-500/40 hover:border-
                blue-500 rounded-2xl p-6 shadow-lg transition duration-200 flex items-start
                gap-4 group"
          &gt;
            &lt;div className="p-3 bg-blue-600/20 text-blue-400 rounded-xl text-3xl group-
                hover:scale-110 transition-transform"&gt;
              ■
            &lt;/div&gt;
            &lt;div&gt;
              &lt;h3 className="text-lg font-bold text-white group-hover:text-blue-400
                  transition-colors"&gt;
                Upload Class Material (PDF)
              &lt;/h3&gt;
              &lt;p className="text-xs text-gray-400 mt-1"&gt;
                Upload lecture notes, practice sheets, and Irodov problem sets for students to
                    view &amp;amp; download.
              &lt;/p&gt;
              &lt;span className="inline-block mt-3 text-xs font-semibold text-blue-400"&gt;
                Manage PDFs →
              &lt;/span&gt;
            &lt;/div&gt;
          &lt;/Link&gt;

          {/* Card 2: Registered Student Database */}
          &lt;Link
            to="/owner/students"
            className="bg-gray-800 hover:bg-gray-750 border border-emerald-500/40
                hover:border-emerald-500 rounded-2xl p-6 shadow-lg transition duration-200
                flex items-start gap-4 group"
          &gt;
            &lt;div className="p-3 bg-emerald-600/20 text-emerald-400 rounded-xl text-3xl group-
                hover:scale-110 transition-transform"&gt;
              ■
            &lt;/div&gt;
            &lt;div&gt;
              &lt;h3 className="text-lg font-bold text-white group-hover:text-emerald-400
```

```
                  transition-colors"&gt;
                Registered Students Database
              &lt;/h3&gt;
              &lt;p className="text-xs text-gray-400 mt-1"&gt;
                View all students registered on the website portal, their emails, phone
                    numbers, and class grades.
              &lt;/p&gt;
              &lt;span className="inline-block mt-3 text-xs font-semibold text-emerald-400"&gt;
                View Student Database →
              &lt;/span&gt;
            &lt;/div&gt;
          &lt;/Link&gt;

          {/* Card 3: See Course Enrollments */}
          &lt;Link
            to="/seeenroll"
            className="bg-gray-800 hover:bg-gray-750 border border-purple-500/40 hover:border-
                purple-500 rounded-2xl p-6 shadow-lg transition duration-200 flex items-start
                gap-4 group"
          &gt;
            &lt;div className="p-3 bg-purple-600/20 text-purple-400 rounded-xl text-3xl group-
                hover:scale-110 transition-transform"&gt;
              ■
            &lt;/div&gt;
            &lt;div&gt;
              &lt;h3 className="text-lg font-bold text-white group-hover:text-purple-400
                  transition-colors"&gt;
                See Course Enrollments
              &lt;/h3&gt;
              &lt;p className="text-xs text-gray-400 mt-1"&gt;
                Check prospective student enrollments, payment screenshots, and 10th marks.
              &lt;/p&gt;
              &lt;span className="inline-block mt-3 text-xs font-semibold text-purple-400"&gt;
                View Applications →
              &lt;/span&gt;
            &lt;/div&gt;
          &lt;/Link&gt;

          {/* Card 4: Upload Photo */}
          &lt;Link
            to="/ownerupload"
            className="bg-gray-800 hover:bg-gray-750 border border-amber-500/40 hover:border-
                amber-500 rounded-2xl p-6 shadow-lg transition duration-200 flex items-start
                gap-4 group"
          &gt;
            &lt;div className="p-3 bg-amber-600/20 text-amber-400 rounded-xl text-3xl group-
                hover:scale-110 transition-transform"&gt;
              ■■
            &lt;/div&gt;
            &lt;div&gt;
              &lt;h3 className="text-lg font-bold text-white group-hover:text-amber-400
                  transition-colors"&gt;
                Upload Gallery Photo
              &lt;/h3&gt;
              &lt;p className="text-xs text-gray-400 mt-1"&gt;
                Upload classroom pictures, student achievements, and institute events to the
                    homepage gallery.
              &lt;/p&gt;
              &lt;span className="inline-block mt-3 text-xs font-semibold text-amber-400"&gt;
                Upload Photos →
              &lt;/span&gt;
            &lt;/div&gt;
          &lt;/Link&gt;
        &lt;/div&gt;
      &lt;/div&gt;
    &lt;/div&gt;
  );
}

export default Owner;
```

■ Frontend/src/components/Enroll.jsx

Purpose: *Course enrollment form: displays UPI QR code, accepts payment screenshot and 10th-grade percentage, and enforces student authentication before allowing access.*

```
import React, { useEffect, useState } from 'react';
import { useForm } from "react-hook-form";
import axios from "axios";
import { useNavigate } from 'react-router-dom';
```

```
import { API_BASE_URL } from '../utils/api';

function Enroll() {
  const navigate = useNavigate();
  const [student, setStudent] = useState(() =&gt; {
    const saved = localStorage.getItem('studentUser');
    return saved ? JSON.parse(saved) : null;
  });

  useEffect(() =&gt; {
    const checkAuth = () =&gt; {
      const saved = localStorage.getItem('studentUser');
      setStudent(saved ? JSON.parse(saved) : null);
    };
    window.addEventListener('studentAuthChange', checkAuth);
    return () =&gt; window.removeEventListener('studentAuthChange', checkAuth);
  }, []);

  const { register, handleSubmit, formState: { errors } } = useForm();

  const onSubmit = async (data) =&gt; {
    const formData = new FormData();
    formData.append("name", data.Name);
    formData.append("fatherName", data.FatherName);
    formData.append("school", data.School);
    formData.append("address", data.Adress);
    formData.append("number", data.Number);
    formData.append("mark", data.Mark);
    formData.append("image", data.Image[0]);

    try {
      await axios.post(`${API_BASE_URL}/api/enroll`, formData, {
        headers: { "Content-Type": "multipart/form-data" }
      });
      navigate("/success");
    } catch (error) {
      console.error("Error submitting form:", error);
      alert("Submission failed");
    }
  };

  // If student is not logged in or registered, show login requirement prompt
  if (!student) {
    return (
      &lt;div className="flex flex-col items-center justify-center min-h-[65vh] bg-amber-50 mt-15
          px-4 text-center"&gt;
        &lt;div className="bg-white p-8 rounded-2xl shadow-xl max-w-md w-full border border-
            gray-200 my-10"&gt;
          &lt;div className="text-5xl mb-4"&gt;■&lt;/div&gt;
          &lt;h2 className="text-2xl font-bold text-gray-800 mb-2"&gt;Student Login Required&lt;/h2&gt;
          &lt;p className="text-sm text-gray-600 mb-6"&gt;
            The course enrollment form is only accessible to registered students. Please log
                in or create an account to proceed with your enrollment.
          &lt;/p&gt;
          &lt;button
            type="button"
            onClick={() =&gt; document.getElementById('my_modal_3')?.showModal()}
            className="w-full py-3 bg-blue-600 hover:bg-blue-700 text-white font-semibold
                rounded-xl shadow cursor-pointer transition duration-150"
          &gt;
            Login / Register to Enroll
          &lt;/button&gt;
        &lt;/div&gt;
      &lt;/div&gt;
    );
  }

  return (
    &lt;div className="flex flex-col items-center text-black bg-amber-50 mt-15 py-8"&gt;
      &lt;div className="text-center mt-6 mb-2 text-2xl font-semibold underline underline-
          offset-4 text-blue-800"&gt;
        Course Enrollment Form
      &lt;/div&gt;
      &lt;p className="text-xs text-gray-600 mb-6"&gt;
        Logged in as &lt;span className="font-semibold text-blue-700"&gt;{student.fullname}&lt;/span&gt;
            ({student.email})
      &lt;/p&gt;

      &lt;form onSubmit={handleSubmit(onSubmit)} encType="multipart/form-data"
          className="space-y-5 bg-white p-6 md:p-8 rounded-2xl shadow-md border border-
          gray-200 w-full max-w-md"&gt;
        &lt;div className="space-y-2"&gt;
```

```
          &lt;span&gt;Enter Your Name&lt;/span&gt;&lt;br /&gt;
          &lt;input
            {...register("Name", { required: true })}
            defaultValue={student.fullname || ''}
            placeholder='Enter Your Name'
            className='w-full px-3 py-2 bg-gray-100 border border-gray-300 rounded-lg text-sm'
          /&gt;
          {errors.Name &amp;&amp; &lt;span className='text-sm text-red-500'&gt;Name is required&lt;/span&gt;}
        &lt;/div&gt;

        &lt;div className="space-y-2"&gt;
          &lt;span&gt;Enter Your Father's Name&lt;/span&gt;&lt;br /&gt;
          &lt;input
            {...register("FatherName", { required: true })}
            placeholder='Enter Your Father Name'
            className='w-full px-3 py-2 bg-gray-100 border border-gray-300 rounded-lg text-sm'
          /&gt;
          {errors.FatherName &amp;&amp; &lt;span className='text-sm text-red-500'&gt;Father's Name is
              required&lt;/span&gt;}
        &lt;/div&gt;

        &lt;div className="space-y-2"&gt;
          &lt;span&gt;Enter Your School Name&lt;/span&gt;&lt;br /&gt;
          &lt;input
            {...register("School", { required: true })}
            placeholder='Enter Your School Name'
            className='w-full px-3 py-2 bg-gray-100 border border-gray-300 rounded-lg text-sm'
          /&gt;
          {errors.School &amp;&amp; &lt;span className='text-sm text-red-500'&gt;School is required&lt;/span&gt;}
        &lt;/div&gt;

        &lt;div className="space-y-2"&gt;
          &lt;span&gt;Enter Your Address&lt;/span&gt;&lt;br /&gt;
          &lt;input
            {...register("Adress", { required: true })}
            placeholder='Enter Your Address'
            className='w-full px-3 py-2 bg-gray-100 border border-gray-300 rounded-lg text-sm'
          /&gt;
          {errors.Adress &amp;&amp; &lt;span className='text-sm text-red-500'&gt;Address is required&lt;/span&gt;}
        &lt;/div&gt;

        &lt;div className="space-y-2"&gt;
          &lt;span&gt;Enter Your Phone Number&lt;/span&gt;&lt;br /&gt;
          &lt;input
            {...register("Number", {
              required: true,
              pattern: /^[0-9]{10}$/
            })}
            defaultValue={student.phone || ''}
            placeholder='Enter 10-digit Phone Number'
            className='w-full px-3 py-2 bg-gray-100 border border-gray-300 rounded-lg text-sm'
          /&gt;
          {errors.Number &amp;&amp; &lt;span className='text-sm text-red-500'&gt;Valid 10-digit number
              required&lt;/span&gt;}
        &lt;/div&gt;

        &lt;div className="space-y-2"&gt;
          &lt;span&gt;Enter Your 10th Mark in %&lt;/span&gt;&lt;br /&gt;
          &lt;input
            {...register("Mark", { required: true })}
            placeholder='Enter Your 10th Mark in %'
            className='w-full px-3 py-2 bg-gray-100 border border-gray-300 rounded-lg text-sm'
          /&gt;
          {errors.Mark &amp;&amp; &lt;span className='text-sm text-red-500'&gt;Mark is required&lt;/span&gt;}
        &lt;/div&gt;

        &lt;div className="space-y-2 border-t border-gray-200 pt-4"&gt;
          &lt;span className="font-semibold text-gray-700"&gt;Pay Rs 1000 to +91 8249687189 &lt;br /&gt;
              (Jagannath Dutta)&lt;/span&gt;&lt;br /&gt;
          &lt;img src="/qr.jpg" className='h-36 rounded-lg border my-2' alt="UPI QR Code" /&gt;
        &lt;/div&gt;

        &lt;div className="space-y-2"&gt;
          &lt;span&gt;Upload Your Payment Screenshot&lt;/span&gt;&lt;br /&gt;
          &lt;input
            {...register("Image", { required: true })}
            type="file"
            accept="image/*"
            className='w-full px-3 py-2 bg-gray-100 border border-gray-300 rounded-lg text-sm
                file:mr-3 file:py-1 file:px-3 file:rounded-md file:border-0 file:bg-blue-600
                file:text-white file:text-xs'
          /&gt;
```

```
169 |           {errors.Image &amp;&amp; &lt;span className='text-sm text-red-500'&gt;Payment screenshot is
170 |               required&lt;/span&gt;}
171 |         &lt;/div&gt;
172 |
173 |         &lt;div className="pt-2"&gt;
174 |           &lt;button type="submit" className='w-full py-3 bg-blue-600 hover:bg-blue-700 text-
175 |               white font-semibold rounded-xl shadow cursor-pointer transition'&gt;
176 |             Submit Enrollment
177 |           &lt;/button&gt;
178 |         &lt;/div&gt;
179 |       &lt;/form&gt;
180 |     &lt;/div&gt;
181 |   );
182 | }
183 |
184 | export default Enroll;
```

■ Frontend/src/components/ViewEnrollments.jsx

Purpose: *Admin enrollment review screen: displays student applications alongside decoded base64 payment screenshots and academic records.*

```
  1 | import React, { useEffect, useState } from "react";
  2 | import axios from "axios";
  3 |
  4 | const ViewEnrollments = () =&gt; {
  5 |   const [data, setData] = useState([]);
  6 |
  7 |   useEffect(() =&gt; {
  8 |     axios
  9 |       .get("https://irodovbackend.onrender.com/api/enroll")
 10 |       .then((res) =&gt; setData(res.data))
 11 |       .catch((err) =&gt; console.error("Failed to fetch:", err));
 12 |   }, []);
 13 |
 14 |   return (
 15 |     &lt;div style={{ padding: "20px" }}&gt;
 16 |       &lt;h2 className="text-3xl text-center text-bold underline "&gt;All Enrollments&lt;/h2&gt;
 17 |       {data.map((item) =&gt; (
 18 |         &lt;div
 19 |           key={item._id}
 20 |           style={{
 21 |             display: "flex",
 22 |             border: "1px solid gray",
 23 |             margin: "10px 0",
 24 |             padding: "15px",
 25 |             borderRadius: "8px",
 26 |             alignItems: "center",
 27 |             justifyContent: "space-between",
 28 |           }}
 29 |         &gt;
 30 |           &lt;div style={{ flex: 1 }}&gt;
 31 |             &lt;p&gt;&lt;strong&gt;Name:&lt;/strong&gt; {item.name}&lt;/p&gt;
 32 |             &lt;p&gt;&lt;strong&gt;Father's Name:&lt;/strong&gt; {item.fatherName}&lt;/p&gt;
 33 |             &lt;p&gt;&lt;strong&gt;School:&lt;/strong&gt; {item.school}&lt;/p&gt;
 34 |             &lt;p&gt;&lt;strong&gt;Address:&lt;/strong&gt; {item.address}&lt;/p&gt;
 35 |             &lt;p&gt;&lt;strong&gt;Number:&lt;/strong&gt; {item.number}&lt;/p&gt;
 36 |             &lt;p&gt;&lt;strong&gt;Mark:&lt;/strong&gt; {item.mark}&lt;/p&gt;
 37 |           &lt;/div&gt;
 38 |           &lt;div style={{ flex: "0 0 200px", marginLeft: "20px" }}&gt;
 39 |             &lt;img
 40 |               src={item.image}
 41 |               alt="Uploaded"
 42 |               style={{ width: "100%", borderRadius: "5px", objectFit: "cover" }}
 43 |             /&gt;
 44 |           &lt;/div&gt;
 45 |         &lt;/div&gt;
 46 |       ))}
 47 |     &lt;/div&gt;
 48 |   );
 49 | };
 50 |
 51 | export default ViewEnrollments;
```

■ Frontend/src/components/OwnerUpload.jsx

Purpose: *Admin photo upload screen: handles uploading institute achievement and event images into the database.*

```
  1 | import React from "react";
  2 | import axios from "axios";
```

```
import { useForm } from "react-hook-form";

function OwnerUpload() {
  const { register, handleSubmit } = useForm();

  const onSubmit = async (data) =&gt; {
    const formData = new FormData();
    formData.append("name", data.name);
    formData.append("image", data.image[0]);

    try {
      await axios.post("https://irodovbackend.onrender.com/upload", formData);
      alert("Uploaded successfully!");
    } catch (err) {
      console.error(err);
      alert("Upload failed");
    }
  };

  return (
  &lt;div className="flex justify-center items-center min-h-screen bg-gray-800"&gt;
  &lt;form
    onSubmit={handleSubmit(onSubmit)}
    className="bg-white shadow-md rounded-xl p-6 w-full max-w-md text-center space-y-4"
  &gt;
    &lt;h2 className="text-2xl font-semibold text-gray-800 mb-4"&gt;Upload Image&lt;/h2&gt;

    &lt;input
      {...register("name")}
      placeholder="Image name"
      className="w-full px-4 py-2 border  rounded-md focus:outline-none focus:ring-2
          focus:ring-blue-500 text-black"
    /&gt;&lt;br /&gt;

    &lt;input
      {...register("image")}
      type="file"
      className="w-full text-sm text-gray-700"
    /&gt;&lt;br /&gt;

    &lt;button
      type="submit"
      className="w-full bg-blue-500 hover:bg-blue-600 text-white font-medium py-2 px-4
          rounded-md transition duration-200"
    &gt;
      Upload
    &lt;/button&gt;
  &lt;/form&gt;
&lt;/div&gt;

  );
}

export default OwnerUpload;
```

■ Frontend/src/components/Banner.jsx

Purpose: *Hero section banner: displays institute banner graphics with a smart 'Enroll now' link (if logged in) or 'Login to Enroll' trigger.*

```
import React, { useEffect, useState } from 'react';

function Banner() {
  const [student, setStudent] = useState(null);

  useEffect(() =&gt; {
    const loadStudent = () =&gt; {
      const saved = localStorage.getItem('studentUser');
      setStudent(saved ? JSON.parse(saved) : null);
    };
    loadStudent();
    window.addEventListener('studentAuthChange', loadStudent);
    return () =&gt; window.removeEventListener('studentAuthChange', loadStudent);
  }, []);

  return (
    &lt;&gt;
      &lt;div&gt;
        &lt;img className="w-full" src="/Banner.png" alt="Irodov Plus Classes Banner" /&gt;
        &lt;div className="text-center mt-7"&gt;
          &lt;h1 className="text-xl sm:text-3xl flex flex-wrap items-center justify-center
              gap-3"&gt;
```

```
            &lt;span&gt;Want to join us?&lt;/span&gt;
            {student ? (
              &lt;a
                href="/enroll"
                className="bg-red-500 sm:text-2xl rounded-lg p-2 px-4 hover:bg-red-700 text-
                    white font-semibold transition"
              &gt;
                Enroll now
              &lt;/a&gt;
            ) : (
              &lt;button
                type="button"
                onClick={() =&gt; document.getElementById('my_modal_3')?.showModal()}
                className="bg-blue-600 sm:text-2xl rounded-lg p-2 px-4 hover:bg-blue-800 text-
                    white font-semibold cursor-pointer transition shadow"
              &gt;
                Login to Enroll
              &lt;/button&gt;
            )}
          &lt;/h1&gt;
        &lt;/div&gt;
      &lt;/div&gt;
    &lt;/&gt;
  );
}

export default Banner;
```

■ Frontend/src/components/Footer.jsx

Purpose: *Global site footer: displays copyright, contact links, social icons, and login-aware enrollment navigation.*

```
import React, { useEffect, useState } from 'react';

function Footer() {
  const [student, setStudent] = useState(null);

  useEffect(() =&gt; {
    const loadStudent = () =&gt; {
      const saved = localStorage.getItem('studentUser');
      setStudent(saved ? JSON.parse(saved) : null);
    };
    loadStudent();
    window.addEventListener('studentAuthChange', loadStudent);
    return () =&gt; window.removeEventListener('studentAuthChange', loadStudent);
  }, []);

  return (
    &lt;div className='mt-10'&gt;
      &lt;footer className="footer footer-horizontal footer-center bg-base-200 text-base-content
          rounded p-10"&gt;
        &lt;nav className="grid grid-flow-col gap-4"&gt;
          &lt;a href='/resources' className="link link-hover"&gt;Study Material&lt;/a&gt;
          &lt;a href='/result' className="link link-hover"&gt;Result&lt;/a&gt;
          {student &amp;&amp; &lt;a href='/enroll' className="link link-hover"&gt;Enroll&lt;/a&gt;}
          &lt;a href='/about' className="link link-hover"&gt;About us&lt;/a&gt;
          &lt;a href='/contact' className="link link-hover"&gt;Contact&lt;/a&gt;
        &lt;/nav&gt;
        &lt;nav&gt;
          &lt;div className="grid grid-flow-col gap-4"&gt;
            &lt;a&gt;
              &lt;svg
                xmlns="http://www.w3.org/2000/svg"
                width="24"
                height="24"
                viewBox="0 0 24 24"
                className="fill-current"&gt;
                &lt;path
                  d="M24 4.557c-.883.392-1.832.656-2.828.775 1.017-.609 1.798-1.574
                      2.165-2.724-.951.564-2.005.974-3.127
                      1.195-.897-.957-2.178-1.555-3.594-1.555-3.179 0-5.515 2.966-4.797
                      6.045-4.091-.205-7.719-2.165-10.148-5.144-1.29 2.213-.669 5.108 1.523
                      6.574-.806-.026-1.566-.247-2.229-.616-.054 2.281 1.581 4.415 3.949
                      4.89-.693.188-1.452.232-2.224.084.626 1.956 2.444 3.379 4.6 3.419-2.07
                      1.623-4.678 2.348-7.29 2.04 2.179 1.397 4.768 2.212 7.548 2.212 9.142 0
                      14.307-7.721 13.995-14.646.962-.695 1.797-1.562 2.457-2.549z"&gt;&lt;/path&gt;
              &lt;/svg&gt;
            &lt;/a&gt;
            &lt;a href='https://www.youtube.com/@jagannathgurujiphysicsclas3268/featured'&gt;
              &lt;svg
                xmlns="http://www.w3.org/2000/svg"
```

```
                width="24"
                height="24"
                viewBox="0 0 24 24"
                className="fill-current"&gt;
                &lt;path
                  d="M19.615 3.184c-3.604-.246-11.631-.245-15.23 0-3.897.266-4.356 2.62-4.385
                      8.816.029 6.185.484 8.549 4.385 8.816 3.6.245 11.626.246 15.23 0
                      3.897-.266 4.356-2.62
                      4.385-8.816-.029-6.185-.484-8.549-4.385-8.816zm-10.615 12.816v-8l8
                      3.993-8 4.007z"&gt;&lt;/path&gt;
              &lt;/svg&gt;
            &lt;/a&gt;
            &lt;a href='https://www.facebook.com/jagan.dutta?mibextid=rS40aB7S9Ucbxw6v'&gt;
              &lt;svg
                xmlns="http://www.w3.org/2000/svg"
                width="24"
                height="24"
                viewBox="0 0 24 24"
                className="fill-current"&gt;
                &lt;path
                  d="M9 8h-3v4h3v12h5v-12h3.642l.358-4h-4v-1.667c0-.955.192-1.333
                      1.115-1.333h2.885v-5h-3.808c-3.596 0-5.192 1.583-5.192
                      4.615v3.385z"&gt;&lt;/path&gt;
              &lt;/svg&gt;
            &lt;/a&gt;
          &lt;/div&gt;
        &lt;/nav&gt;
        &lt;aside&gt;
          &lt;p&gt;Copyright © {new Date().getFullYear()} - All right reserved by Irodov plus
              Classes Baripada&lt;/p&gt;
        &lt;/aside&gt;
      &lt;/footer&gt;
    &lt;/div&gt;
  );
}

export default Footer;
```

■ Frontend/src/components/PhotoGallery.jsx

Purpose: *Public photo gallery: fetches and renders classroom and event images stored in the MongoDB Image collection.*

```
import React, { useEffect, useState } from "react";
import axios from "axios";
import { API_BASE_URL } from "../utils/api";

function PhotoGallery() {
  const [images, setImages] = useState([]);

  useEffect(() =&gt; {
    axios
      .get(`${API_BASE_URL}/images`)
      .then((res) =&gt; {
        setImages(res.data);
      })
      .catch((err) =&gt; {
        console.log("Gallery fetch error:", err);
      });
  }, []);

  return (
    &lt;div className="mt-20 sm:mx-20"&gt;
      &lt;h1 className="text-bold text-center text-3xl"&gt;Photo Gallery&lt;/h1&gt;

      &lt;div className="flex flex-wrap gap-4 mx-10 sm:p-4"&gt;
        {images.map((img, index) =&gt; (
          &lt;div key={index}&gt;
            &lt;img
              src={`data:${img.image.contentType};base64,${btoa(
                new Uint8Array(img.image.data.data).reduce(
                  (data, byte) =&gt; data + String.fromCharCode(byte),
                  ""
                )
              )}`}
              alt={img.name || "gallery"}
              className="w-50 h-50 object-cover border rounded"
            /&gt;
          &lt;/div&gt;
        ))}
      &lt;/div&gt;
    &lt;/div&gt;
```

```
  );
}

export default PhotoGallery;
```

■ Frontend/src/components/Weprovide.jsx

Purpose: *Academic offerings showcase: cards detailing special physics batches, foundation courses, and problem-solving workshops.*

```
import React, { useEffect, useState } from 'react';

function Weprovide() {
  const [provide, setProvide] = useState([]);

  useEffect(() =&gt; {
    fetch('/provide.json')
      .then((res) =&gt; res.json())
      .then((data) =&gt; setProvide(data))
      .catch((err) =&gt; console.log(err));
  }, []);

  return (
    &lt;&gt;
      &lt;div className="mt-15"&gt;
        &lt;h1 className="text-3xl font-bold text-center mb-8 mx-auto"&gt;We Provide&lt;/h1&gt;

        &lt;div className="flex flex-wrap mx-5 gap-4 sm:gap-6 justify-center"&gt;
          {provide.map((item) =&gt; (
            &lt;div
              className="card bg-amber-50 text-black w-40 sm:w-80 md:w-96 min-h-[320px]
                  shadow-sm"
              key={item.id}
            &gt;
              &lt;figure className="px-6 pt-6"&gt;
                &lt;img
                  src={item.image}
                  alt={item.title}
                  className="w-full h-32 sm:h-48 object-contain rounded-xl"
                /&gt;
              &lt;/figure&gt;

              &lt;div className="card-body items-center text-center pt-4"&gt;
                &lt;h2 className="card-title text-lg sm:text-2xl"&gt;{item.name}&lt;/h2&gt;
                &lt;p className="text-sm sm:text-xl"&gt;{item.title}&lt;/p&gt;
              &lt;/div&gt;
            &lt;/div&gt;
          ))}
        &lt;/div&gt;
      &lt;/div&gt;
    &lt;/&gt;
  );
}

export default Weprovide;
```

■ Frontend/src/components/Academics.jsx

Purpose: *Academic curriculum section: outlines teaching methodology, syllabus pacing, and test series structure.*

```
import React from 'react'

function Academics() {
  return (



        &lt;&gt;
            &lt;div className=''&gt;
                &lt;h2 className="text-center text-1xl lg:text-3xl font-bold py-2 mt-5"&gt;
                    A Coaching Trusted by Students
                &lt;/h2&gt;
                &lt;h5 className="text-center lg:text-2xl ml-4 mb-5 sm:ml-10"&gt;
                    Irodov Classes aims to transform not just through words, but provide
                        results with numbers!
                &lt;/h5&gt;
            &lt;/div&gt;
            &lt;div className="grid grid-cols-2 lg:grid-cols-4 gap-4 items-center justify-center
                bg-amber-50 text-black "&gt;
                &lt;div className="rounded-sm bg-sky-300 mx-7 h-40 py-10 my-5"&gt;
```

```
                    &lt;h2 className="text-center text-2xl sm:text-5xl font-bold"&gt;10+&lt;/h2&gt;
                    &lt;h4 className="text-center text-1xl sm:text-2xl px-10"&gt;Jee Qualified&lt;/h4&gt;
                &lt;/div&gt;
                &lt;div className="rounded-sm bg-pink-300 mx-7 h-40 py-10 my-5"&gt;
                    &lt;h2 className="text-center text-2xl sm:text-5xl font-bold"&gt;5+&lt;/h2&gt;
                    &lt;h4 className="text-center text-1xl sm:text-2xl px-10"&gt;NEET Qualified&lt;/h4&gt;
                &lt;/div&gt;
                &lt;div className="rounded-sm bg-red-300 mx-7 h-40 py-10 my-5"&gt;
                    &lt;h2 className="text-center text-2xl sm:text-5xl font-bold"&gt;20+&lt;/h2&gt;
                    &lt;h4 className="text-center text-1xl sm:text-2xl px-10"&gt;90% in Board&lt;/h4&gt;
                &lt;/div&gt;
                &lt;div className="rounded-sm bg-fuchsia-300 mx-7 h-40 py-10 my-5"&gt;
                    &lt;h2 className="text-center text-2xl sm:text-5xl font-bold"&gt;5+&lt;/h2&gt;
                    &lt;h4 className="text-center text-1xl sm:text-2xl px-10"&gt;OUAT Qualified&lt;/h4&gt;
                &lt;/div&gt;


    &lt;/div&gt;
    &lt;/&gt;
  )
}

export default Academics;
```

■ Frontend/src/components/About.jsx

Purpose: *About page component: biography of JD Sir, teaching philosophy, and coaching history.*

```
import React from 'react'

function About() {
  return (
   &lt;&gt;


    &lt;div style={{ paddingTop: "84px" }} className="p-6  md:p-10 bg-white text-gray-800
        max-w-5xl mx-auto"&gt;
      &lt;h1 className="text-4xl font-bold mb-6 text-blue-600"&gt;About Us&lt;/h1&gt;
      &lt;p className="text-lg mb-6"&gt;
        &lt;strong&gt;Irodov Plus Classes&lt;/strong&gt; is a trusted{" "}
        &lt;strong&gt;offline education institute based in Baripada&lt;/strong&gt;,
        dedicated to delivering high-quality, result-oriented coaching for{" "}
        &lt;strong&gt;Classes 11 and 12 (CBSE Board)&lt;/strong&gt; along with specialized
        preparation for &lt;strong&gt;JEE&lt;/strong&gt; and &lt;strong&gt;NEET&lt;/strong&gt; aspirants.
        Our goal is to make learning focused, accessible, and personal for every
        student who walks through our doors.
      &lt;/p&gt;

      &lt;p className="text-lg mb-6"&gt;
        We believe in building a strong academic foundation through{" "}
        &lt;strong&gt;face-to-face classroom learning&lt;/strong&gt;, one-on-one doubt
        clearing, regular assessments, and consistent motivation. With
        experienced faculty and a structured academic plan, we ensure our
        students are not only well-prepared for board exams but also competitive
        enough to crack national-level entrance exams.
      &lt;/p&gt;

      &lt;h2 className="text-3xl font-semibold mt-10 mb-4 text-blue-500"&gt;
        Why Choose Us
      &lt;/h2&gt;
      &lt;p className="text-lg mb-6"&gt;
        We stand out because we combine{" "}
        &lt;strong&gt;classroom excellence with personal mentorship&lt;/strong&gt;. Our
        faculty members are highly qualified and have mentored thousands of
        students towards academic and competitive success. At our Baripada
        center, students get the benefits of small batch sizes, regular
        parent-teacher interactions, personalized progress tracking, and a
        disciplined yet friendly learning atmosphere.
      &lt;/p&gt;
      &lt;p className="text-lg mb-6"&gt;
        We are not just a coaching center—we are a{" "}
        &lt;strong&gt;launchpad for aspiring engineers and doctors&lt;/strong&gt; from
        Odisha .
      &lt;/p&gt;

      &lt;h2 className="text-3xl font-semibold mt-10 mb-4 text-blue-500"&gt;
        Our Focus Areas
      &lt;/h2&gt;
      &lt;ul className="list-disc list-inside text-lg mb-6 space-y-2"&gt;
        &lt;li&gt;
          &lt;strong&gt;CBSE Board Coaching (Classes 11 &amp;amp; 12)&lt;/strong&gt;: Detailed,
```

```
          syllabus-oriented teaching in &lt;strong&gt;Physics, Chemistry, Mathematics ,English &amp;
              Biology&lt;/strong&gt; for excellent board performance.
        &lt;/li&gt;
        &lt;li&gt;
          &lt;strong&gt;JEE &amp; NEET Preparation&lt;/strong&gt;: Expertly crafted courses designed to
              strengthen concepts, enhance problem-solving, and boost exam confidence.
        &lt;/li&gt;
        &lt;li&gt;
          &lt;strong&gt;Regular Assessments &amp; Feedback&lt;/strong&gt;: Weekly and monthly tests with
              detailed analysis and reports to help students improve continuously.
        &lt;/li&gt;
        &lt;li&gt;
          &lt;strong&gt;Doubt-Clearing Sessions&lt;/strong&gt;: Daily interactive sessions for concept
              clarity and personal attention.
        &lt;/li&gt;
        &lt;li&gt;
          &lt;strong&gt;Printed Notes &amp; Study Material&lt;/strong&gt;: Well-curated handouts, practice
              sheets, and revision modules tailored to current exam patterns.
        &lt;/li&gt;
      &lt;/ul&gt;

      &lt;h2 className="text-3xl font-semibold mt-10 mb-4 text-blue-500"&gt;
        What Makes Us Different
      &lt;/h2&gt;
      &lt;ul className="list-disc list-inside text-lg mb-6 space-y-2"&gt;
        &lt;li&gt;
          &lt;strong&gt;Offline &amp; Interactive&lt;/strong&gt;: We believe the classroom is still the most
              effective place to learn. Our physical presence ensures greater focus,
              discipline, and peer learning.
        &lt;/li&gt;
        &lt;li&gt;
          &lt;strong&gt;Local Presence, National-Level Standards&lt;/strong&gt;: Bringing metro-quality
              education to Baripada with a focus on &lt;strong&gt;affordability&lt;/strong&gt; and
              &lt;strong&gt;accessibility&lt;/strong&gt;.
        &lt;/li&gt;
        &lt;li&gt;
          &lt;strong&gt;Parent Connect&lt;/strong&gt;: Regular progress updates and PTMs to keep families
              involved in the student’s journey.
        &lt;/li&gt;
        &lt;li&gt;
          &lt;strong&gt;Holistic Support&lt;/strong&gt;: Academic, emotional, and motivational support to
              help students manage stress and stay goal-oriented.
        &lt;/li&gt;
      &lt;/ul&gt;

      &lt;p className="text-lg mb-8"&gt;
        Whether you’re aiming to &lt;strong&gt;top your board exams&lt;/strong&gt; or secure a seat in{"
            "}
        &lt;strong&gt;IITs or top medical colleges&lt;/strong&gt;, &lt;strong&gt;Irodov Plus Classes&lt;/strong&gt; in
            Baripada is here to guide you every step of the way.
      &lt;/p&gt;


    &lt;/div&gt;


   &lt;/&gt;
  )
}

export default About
```

■ Frontend/src/components/Contact.jsx

Purpose: *Contact details component: phone numbers, coaching address in Baripada, and Google Maps embed.*

```
import React from 'react'

function Contact() {
  return (
   &lt;&gt;

    &lt;div  style={{ paddingTop: "154px" }} className="bg-[#fdf8f4] min-h-screen p-8 flex flex-
        col items-center mb-0"&gt;
      &lt;h1 className="text-4xl font-bold text-black mb-12"&gt;Contact us&lt;/h1&gt;

      &lt;div className="flex flex-col md:flex-row gap-8 w-full max-w-4xl justify-center"&gt;
        {/* Contact Sales Card */}
        &lt;div className="bg-white p-6 rounded-lg shadow-md flex-1 text-center"&gt;
          &lt;div className="text-4xl mb-4 flex items-center justify-center"&gt;
            &lt;img className='h-32' src="map.png" alt="" /&gt;
```

```
          &lt;/div&gt;
          &lt;h2 className="text-black text-xl font-semibold mb-2"&gt;Irodov Plus Classes&lt;/h2&gt;
          &lt;p className="text-blue-600 mb-6"&gt;
At - kolho sahi, debendrapur road, near autonomous college, baripada, 757003         &lt;/p&gt;

        &lt;/div&gt;

        {/* Help Center Card */}
        &lt;div className="bg-white p-6 rounded-lg shadow-md flex-1 text-center"&gt;
          &lt;div className="text-4xl mb-4 flex items-center justify-center"&gt;
&lt;img src="call.png" className='h-32' alt="" /&gt;
          &lt;/div&gt;
          &lt;h2 className="text-xl text-black font-semibold mb-2"&gt;Ph.: +91 85960 64667&lt;/h2&gt;
          &lt;p className="text-blue-600 mb-6"&gt;
       Email: rjdutta.91@gmail.com
          &lt;/p&gt;

        &lt;/div&gt;
      &lt;/div&gt;
    &lt;/div&gt;



   &lt;/&gt;
  )
}

export default Contact
```

■ Frontend/src/components/Filterphoto.jsx

Purpose: *Gallery filtering component allowing users to sort photos by category.*

```
import React, { useEffect, useState } from "react";
import axios from "axios";

function Filterphoto() {
  const [images, setImages] = useState([]);

  useEffect(() =&gt; {
    axios.get("https://irodovbackend.onrender.com/images")
      .then(res =&gt; setImages(res.data))
      .catch(err =&gt; console.log(err));
  }, []);

  return (
    &lt;&gt;
      &lt;div className="mt-20 mx-5 sm:mx-15 "&gt;
        &lt;h1 className="font-bold text-center text-3xl mb-6"&gt;Photo Gallery&lt;/h1&gt;
        &lt;div className="flex flex-wrap gap-4  sm:p-4 "&gt;
          {images.slice(0, 12).map((img, index) =&gt; (
            &lt;div key={index}&gt;
              &lt;img
                src={`data:${img.image.contentType};base64,${btoa(
                  new Uint8Array(img.image.data.data).reduce(
                    (data, byte) =&gt; data + String.fromCharCode(byte),
                    ""
                  )
                )}`}
                alt={img.name}
            className=" w-35 h-35 sm:w-50 sm:h-50 object-cover border rounded"
              /&gt;
            &lt;/div&gt;
          ))}
        &lt;/div&gt;
      &lt;/div&gt;
    &lt;/&gt;
  );
}

export default Filterphoto;
```

■ Frontend/src/components/Photo.jsx

Purpose: *Presentational component rendering an individual gallery photo card.*

```
import React from 'react'

function Photo() {
```

```
  return (
    &lt;div&gt;

    &lt;/div&gt;
  )
}

export default Photo
```

■ Frontend/src/components/Result.jsx

Purpose: *Student results overview component: displays past top ranks and student testimonials.*

```
import React from 'react'
import PhotoGallery from "./PhotoGallery";
import Nit from './Results/Nit';
import Ojee from './Results/Ojee';
import Board from './Results/Board';

function Result() {
  return (
    &lt;&gt;
    &lt;Nit/&gt;
    &lt;Ojee/&gt;
    &lt;Board/&gt;
    &lt;div&gt;

 &lt;PhotoGallery/&gt;

    &lt;/div&gt;
    &lt;/&gt;
  )
}

export default Result
```

■ Frontend/src/components/Slider.jsx

Purpose: *Interactive photo slider component for displaying featured institute banners.*

```
import React from 'react'
import SliderPage from '../components/SliderPage'

function Slider() {
  return (
    &lt;div className=' '&gt;
      &lt;SliderPage/&gt;
    &lt;/div&gt;
  )
}

export default Slider
```

■ Frontend/src/components/SliderPage.jsx

Purpose: *Container page for image slider presentations.*

```
import React, { useEffect, useState } from 'react';
import Nit from './Results/Nit';
import Ojee from './Results/Ojee';
import Board from './Results/Board';
import { AnimatePresence, motion } from 'framer-motion';

const slides = [&lt;Nit /&gt;, &lt;Ojee /&gt;, &lt;Board /&gt;];

export default function SliderPage() {
  const [index, setIndex] = useState(0);

  useEffect(() =&gt; {
    const interval = setInterval(() =&gt; {
      setIndex((prev) =&gt; (prev + 1) % slides.length);
    }, 5000);

    return () =&gt; clearInterval(interval);
  }, []);

  return (
```

```
21 |     &lt;div className="w-full    "&gt;
22 |       &lt;AnimatePresence mode="wait"&gt;
23 |         &lt;motion.div
24 |           key={index}
25 |           initial={{ opacity: 0, x: 100 }}
26 |           animate={{ opacity: 1, x: 0 }}
27 |           exit={{ opacity: 0, x: -100 }}
28 |           transition={{ duration: 0.5 }}
29 |           className="w-full " // Removed 'absolute'
30 |         &gt;
31 |           {slides[index]}
32 |         &lt;/motion.div&gt;
33 |       &lt;/AnimatePresence&gt;
34 |     &lt;/div&gt;
35 |   );
36 | }
```

■ Frontend/src/components/Staffdetails.jsx

Purpose: *Faculty profile component highlighting qualifications and mentoring experience.*

```
 1 | import React from 'react'
 2 |
 3 | function Staffdetails() {
 4 |
 5 |     return (
 6 |       &lt;div className=" p-10 rounded-lg my-5"&gt;
 7 |         &lt;h1 className="text-3xl font-bold text-center"&gt;Institute Information&lt;/h1&gt;
 8 |         &lt;hr className="my-4 border-gray-700" /&gt;
 9 |         &lt;div className="flex flex-col md:flex-row items-center justify-between"&gt;
10 |           &lt;div className="md:w-1/2 flex justify-center"&gt;
11 |             &lt;img src="logo.png" alt="Ranking" className="w-3/4 md:w-1/2" /&gt;
12 |           &lt;/div&gt;
13 |           &lt;div className=" w-90 md:w-1/2 grid grid-cols-2 gap-6 mt-6 md:mt-0"&gt;
14 |             &lt;div className="flex flex-col items-center bg-gray-800 p-4 rounded-lg shadow-lg"&gt;
15 |               &lt;img src="/staff/department.png" alt="Department" className="w-16 h-16 mb-2" /&gt;
16 |               &lt;span className="text-2xl font-bold"&gt;5+&lt;/span&gt;
17 |               &lt;p&gt;Subject&lt;/p&gt;
18 |             &lt;/div&gt;
19 |             &lt;div className="flex flex-col items-center bg-gray-800 p-4 rounded-lg shadow-lg"&gt;
20 |               &lt;img src="/staff/student-icon.png" alt="Students" className="w-16 h-16 mb-2" /&gt;
21 |               &lt;span className="text-2xl font-bold"&gt;100+&lt;/span&gt;
22 |               &lt;p&gt;Students&lt;/p&gt;
23 |             &lt;/div&gt;
24 |             &lt;div className="flex flex-col items-center bg-gray-800 p-4 rounded-lg shadow-lg"&gt;
25 |               &lt;img src="/staff/faculty.png" alt="Faculty" className="w-16 h-16 mb-2" /&gt;
26 |               &lt;span className="text-2xl font-bold"&gt;10+&lt;/span&gt;
27 |               &lt;p&gt;Faculty&lt;/p&gt;
28 |             &lt;/div&gt;
29 |             &lt;div className="flex flex-col items-center bg-gray-800 p-4 rounded-lg shadow-lg"&gt;
30 |               &lt;img src="/staff/staff.png" alt="Staff" className="w-16 h-16 mb-2" /&gt;
31 |               &lt;span className="text-2xl font-bold"&gt;5+&lt;/span&gt;
32 |               &lt;p&gt;Staff&lt;/p&gt;
33 |             &lt;/div&gt;
34 |           &lt;/div&gt;
35 |         &lt;/div&gt;
36 |       &lt;/div&gt;
37 |     );
38 |   }
39 |
40 |
41 |
42 | export default Staffdetails
```

■ Frontend/src/components/Sucess.jsx

Purpose: *Success confirmation screen displayed after a student submits their course enrollment application.*

```
 1 | import React from 'react'
 2 |
 3 |
 4 |
 5 | function Success() {
 6 |   return (
 7 |     &lt;&gt;
 8 |     &lt;div className='flex justify-center items-center h-screen bg-green-100'&gt;
 9 |       &lt;div className='text-center'&gt;
10 |         &lt;h1 className='text-3xl font-bold text-green-700 mb-4'&gt;■ Congratulations!&lt;/h1&gt;
11 |         &lt;p className='text-xl text-gray-800'&gt;Your seat has been successfully confirmed.&lt;/p&gt;
```

```
      &lt;/div&gt;
    &lt;/div&gt;
    &lt;/&gt;
  )
}

export default Success;
```

■ Frontend/src/components/Enrolls/Enrolls.jsx

Purpose: *Page container wrapping Navbar, the Enroll form component, and Footer.*

```
import React from 'react'
import Enroll from '../Enroll'
import Navbar from '../Navbar'
import Footer from '../Footer'
function Enrolls() {
  return (
    &lt;div&gt;
      &lt;Navbar/&gt;
      &lt;Enroll/&gt;
      &lt;Footer/&gt;

    &lt;/div&gt;
  )
}

export default Enrolls
```

■ Frontend/src/components/Results/Board.jsx

Purpose: *CHSE/CBSE 12th Board examination top scorers and distinction list.*

```
import React, { useEffect, useState } from 'react';

function Board() {
  const [provide, setProvide] = useState([]);

  useEffect(() =&gt; {
    fetch('/Class12.json')
      .then((res) =&gt; res.json())
      .then((data) =&gt; setProvide(data));
  }, []);

  return (
    &lt;&gt;
      &lt;div className="mt-15 px-2 py-4 bg-amber-200"&gt;
  &lt;h1 className="text-2xl font-bold text-black text-center mb-2"&gt;
    Our Class 12 Result
  &lt;/h1&gt;
  &lt;div className="flex flex-nowrap overflow-x-auto gap-1 sm:gap-3 px-0 sm:px-2 sm:mx-15"&gt;
    {provide.map((item) =&gt; (
      &lt;div
        key={item.id}
        className="bg-amber-50 text-black w-55 sm:w-70 flex-shrink-0 rounded-lg shadow-md
            hover:shadow-lg transition duration-300 flex flex-col"
      &gt;
        &lt;figure className="px-0 pt-0 sm:px-2 sm:pt-2"&gt;
          &lt;img
            src={item.image}
            alt={item.title}
            className="rounded-md object-cover object-top mt-2 mb-4 w-50 h-70 ml-2  h-14
                sm:h-90 sm:w-63 "
          /&gt;
        &lt;/figure&gt;

      &lt;/div&gt;
    ))}
  &lt;/div&gt;
&lt;/div&gt;
    &lt;/&gt;
  );
}

export default Board;
```

■ Frontend/src/components/Results/Nit.jsx

Purpose: *NIT and JEE Main qualified students achievement showcase.*

```
import React, { useEffect, useState } from 'react';

function Nit() {
  const [provide, setProvide] = useState([]);

  useEffect(() =&gt; {
    fetch('/nit.json')
      .then((res) =&gt; res.json())
      .then((data) =&gt; setProvide(data));
  }, []);

  return (
    &lt;&gt;
      &lt;div className="mt-15 px-2 py-4 bg-amber-200"&gt;
  &lt;h1 className="text-2xl font-bold text-black text-center mb-2"&gt;
    Our JEE Result
  &lt;/h1&gt;
  &lt;div className="flex flex-nowrap overflow-x-auto gap-1 sm:gap-3 px-0 sm:px-2 sm:mx-15"&gt;
    {provide.map((item) =&gt; (
      &lt;div
        key={item.id}
        className="bg-amber-50 text-black w-28 sm:w-64 flex-shrink-0 rounded-lg shadow-md
            hover:shadow-lg transition duration-300 flex flex-col"
      &gt;
        &lt;figure className="px-0 pt-0 sm:px-2 sm:pt-2"&gt;
          &lt;img
            src={item.image}
            alt={item.title}
            className="rounded-md object-cover object-top mt-2 ml-4 w-20  h-14 sm:h-44 sm:w-52
                "
          /&gt;
        &lt;/figure&gt;
        &lt;div className="px-1 py-1 sm:px-4 sm:py-3 text-center"&gt;
          &lt;h2 className="font-semibold text-xs sm:text-xl"&gt;{item.name}&lt;/h2&gt;
          &lt;p className="text-[10px] sm:text-sm text-gray-800"&gt;{item.title}&lt;/p&gt;
          &lt;p className="text-[10px] sm:text-sm text-gray-600"&gt;{item.description}&lt;/p&gt;
        &lt;/div&gt;
      &lt;/div&gt;
    ))}
  &lt;/div&gt;
&lt;/div&gt;
    &lt;/&gt;
  );
}

export default Nit;
```

■ Frontend/src/components/Results/Ojee.jsx

Purpose: *OJEE state rank holders achievement showcase.*

```
import React, { useEffect, useState } from 'react';

function Ojee() {
  const [provide, setProvide] = useState([]);

  useEffect(() =&gt; {
    fetch('/ojee.json')
      .then((res) =&gt; res.json())
      .then((data) =&gt; setProvide(data));
  }, []);

  return (
    &lt;&gt;
      &lt;div className="mt-15 px-2 py-4 bg-amber-200"&gt;
  &lt;h1 className="text-2xl font-bold text-black text-center mb-2"&gt;
    Our OJEE Result
  &lt;/h1&gt;
  &lt;div className="flex flex-nowrap overflow-x-auto gap-1 sm:gap-3 px-0 sm:px-2 sm:mx-15"&gt;
    {provide.map((item) =&gt; (
      &lt;div
        key={item.id}
        className="bg-amber-50 text-black w-45 sm:w-68 flex-shrink-0 rounded-lg shadow-md
            hover:shadow-lg transition duration-300 flex flex-col"
```

```
24 |       &gt;
25 |         &lt;figure className="px-0 pt-0 sm:px-2 sm:pt-2"&gt;
26 |           &lt;img
27 |             src={item.image}
28 |             alt={item.title}
29 |             className="rounded-md object-cover object-top mt-2 mb-4 w-40 h-40 ml-2  h-14
30 |                 sm:h-65 sm:w-60 "
31 |           /&gt;
32 |         &lt;/figure&gt;
33 |
34 |       &lt;/div&gt;
35 |     ))}
36 |   &lt;/div&gt;
37 | &lt;/div&gt;
38 |     &lt;/&gt;
39 |   );
40 | }
41 |
42 | export default Ojee;
```

■ Frontend/src/Home/Home.jsx

Purpose: *Homepage container assembling Banner, We Provide, Gallery, Academics, and Footer.*

```
 1 | import React from 'react'
 2 |
 3 |
 4 | import Navbar from '../components/Navbar'
 5 | import Banner from "../components/Banner"
 6 | import Footer from "../components/Footer"
 7 | import Weprovide from "../components/Weprovide"
 8 | import Academics from "../components/Academics"
 9 | import Staffdetails from '../components/Staffdetails'
10 | import Filterphoto from '../components/Filterphoto'
11 | import Slider from '../components/Slider'
12 | function Home() {
13 |
14 |
15 |   return (
16 |     &lt;&gt;
17 |     &lt;Navbar/&gt;
18 |     &lt;Slider/&gt;
19 |     &lt;Banner/&gt;
20 |     &lt;Weprovide/&gt;
21 |     &lt;Academics/&gt;
22 |     &lt;Staffdetails/&gt;
23 | &lt;Filterphoto/&gt;
24 |     &lt;Footer/&gt;
25 |
26 |     &lt;/&gt;
27 |   )
28 | }
29 |
30 | export default Home
```

■ Frontend/src/Results/Results.jsx

Purpose: *Results page container assembling result categories and rankings.*

```
 1 | import React from 'react'
 2 | import Result from '../components/Result'
 3 | import Navbar from '../components/Navbar'
 4 | import Footer from '../components/Footer'
 5 |
 6 | function Results() {
 7 |   return (
 8 |     &lt;div&gt;
 9 |       &lt;Navbar/&gt;
10 |       &lt;div className='min-h-screen'&gt;
11 |        &lt;Result/&gt;
12 |       &lt;/div&gt;
13 |
14 |       &lt;Footer/&gt;
15 |     &lt;/div&gt;
16 |   )
17 | }
18 |
19 | export default Results
```

■ Frontend/src/Aboutus/Aboutus.jsx

Purpose: *About Us page container assembling institute history and faculty profiles.*

```
 1 | import React from 'react'
 2 | import About from '../components/About'
 3 | import Navbar from '../components/Navbar'
 4 | import Footer from '../components/Footer'
 5 | function Aboutus() {
 6 |   return (
 7 |     &lt;div&gt;
 8 |       &lt;Navbar/&gt;
 9 |       &lt;div className='min-h-screen'&gt;
10 |        &lt;About/&gt;
11 |       &lt;/div&gt;
12 |
13 |       &lt;Footer/&gt;
14 |     &lt;/div&gt;
15 |   )
16 | }
17 |
18 | export default Aboutus
```

■ Frontend/src/Contactus/Contactus.jsx

Purpose: *Contact Us page container assembling contact cards and enquiry forms.*

```
 1 | import React from 'react'
 2 | import Navbar from '../components/Navbar'
 3 | import Contact from '../components/Contact'
 4 | import Footer from '../components/Footer'
 5 | function Contactus() {
 6 |   return (
 7 |     &lt;div&gt;
 8 |       &lt;Navbar/&gt;
 9 |       &lt;div className='min-h-screen'&gt;
10 |        &lt;Contact/&gt;
11 |       &lt;/div&gt;
12 |
13 |       &lt;Footer/&gt;
14 |     &lt;/div&gt;
15 |   )
16 | }
17 |
18 | export default Contactus
```